BOUNDS FOR THE LENGTH OF FINITE PIERCE EXPANSIONS

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ABSTRACT. For integers $a > b > 0$, let $P(a,b)$ count the positive terms of the iteration $x \mapsto a \bmod x$, starting at b. We prove

$$P(a,b) \ll_\varepsilon a^{\alpha_*+\varepsilon}, \qquad \alpha_* = (1310 - \sqrt{655})/4365 = 0.2942513246\ldots.$$

The rational exponents 2201/7480 and 128/435 also follow. We group short segments of the orbit by integer determinants derived from a positive moment model. For each group, the fixed-dividend equations place the possible quotient increments on affine curves. Additional integer invariants give embeddings of higher degree while keeping the coordinates small. A coefficient-uniform curve-counting theorem then bounds the number of segments in each group. Taking powers of the added coordinates and balancing their heights gives the exponent α_*. We also prove several elementary intermediate bounds, a bound for trajectories with few rises in their quotient increments, and positivity and integrality of the associated Hankel determinants. The main result and its curve-counting input are formalized in Lean 4 with Mathlib without proof holes or additional axioms. The Lean proof of a separate logarithmic refinement of the 3/10 bound takes two published analytic estimates as explicit hypotheses. Shallit's conjectured $O((\log a)^2)$ bound remains open.

1. INTRODUCTION

Let $a > b > 0$ be integers. Put $b_1 = b$ and $b_{i+1} = a \bmod b_i$ while $b_i > 0$. The sequence is strictly decreasing until it reaches 0; if $b_{n+1} = 0$ we write $P(a,b) = n$. For example $P(35,22) = 7$, from $22, 13, 9, 8, 3, 2, 1, 0$. The number $P(a,b)$ is the length of the Pierce expansion of b/a [12, 13], since

$$\frac{b}{a} = \frac{1}{q_1} - \frac{1}{q_1 q_2} + \frac{1}{q_1 q_2 q_3} - \cdots, \qquad q_i = \lfloor a/b_i \rfloor.$$

Shallit [11] asked whether

$$P(a,b) = O\big((\log a)^2\big). \tag{1}$$

Shallit [13] proved $P(a,b) \ll a^{1/2}$ by "Archimedean" arguments. Erdős and Shallit [6] combined these with a divisor argument to get $P(a,b) \ll_\varepsilon a^{1/3+\varepsilon}$, and showed that $\max_b P(a,b)/\log a$ does not tend to 0. Chase and Pandey [5] went past the exponent 1/3, proving $P(a,b) \ll_\varepsilon a^{1/3-2/177+\varepsilon}$ with exponential sum estimates (Bourgain's exponent pair), and showed that $\max_b P(a,b) \gg \log a/\log\log a$ for all large a. Baraskar and Vukusic [1] study the related sets of iterated remainders and bound their cardinalities at each fixed iteration depth. Shallit's talk pages [16] report a bound $n^{3/10}$, described as "LLM, unpublished so far". We have not seen that argument. The preliminary 3/10 theorem below has the same exponent up to ε, and we make no claim about how the two arguments are related.

Our strongest bound comes from the joint-invariant curve embedding of Section 15.

Theorem 1.1. *For every $\varepsilon > 0$ there is $C_\varepsilon > 0$, independent of a, b, such that*

$$P(a,b) \le C_\varepsilon a^{(1310-\sqrt{655})/4365+\varepsilon} \qquad (0 < b < a).$$

The rational exponents 2201/7480 *and* 128/435 *also give such uniform bounds.*

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How the main argument fits together. We count orbit terms in a remainder window $[A, 2A)$, then sum over dyadic values of A. For most short segments in a window, both the quotient increments and certain integer determinants are small. We call the collection of determinants a *pattern*. Fixing a pattern reduces the count to a system of polynomial equations in the increments.

The proof has two main stages. Sections 10–13 construct the patterns and bound their number. Section 15 retains the fixed dividend a in the equations, so that each pattern gives finitely many curves. It then adjoins small integer invariants as coordinates. The degree of the resulting curve controls the saving in the point count: larger degree gives a smaller exponent of the coordinate bound. The final step balances the competing coordinate bounds.

The elementary window estimates and the six-step identity are developed first, in Sections 3–7. The short-interval divisor refinement of Section 8, the rise bound of Section 9, and the lattice refinement of Section 14 can be read separately; they are not needed for the joint-invariant bound. Section 16 explains the scope of the formal verification.

The first window bound. We first develop the elementary ingredients and earlier bounds. The first is:

Theorem 1.2. *For every $\varepsilon > 0$ there is a constant C_ε such that*

$$P(a,b) \le C_\varepsilon\, a^{3/10+\varepsilon} \qquad \text{for all integers } 0 < b < a.$$

The proof is elementary. It uses only the division equations $a = q_i b_i + b_{i+1}$ and the divisor bound $\tau(n) \ll_\delta n^\delta$. As usual, the remainders are split into dyadic windows $[A, 2A)$, and we bound the number T of remainders in one window. There are two bounds:

$$T(T-1) \le 4A\,\Delta, \tag{2}$$

$$T^6 A^2 \le 995328\, a^3 \Delta \qquad (T \ge 12), \tag{3}$$

where Δ is the largest value of the divisor function on $[1, 32a^7]$. Bound (2) is the Erdős–Shallit argument. Bound (3) is new. Multiplying (3) by the square of (2) eliminates A and gives $T^{10} \ll a^3\Delta^3$, which is the exponent $3/10$.

The new bound comes from three consecutive steps $x = b_i$, $y = b_{i+1}$, $z = b_{i+2}$, $w = b_{i+3}$. Let d, e, f be the drops, let u, v be the increments of the quotients, and put $D = dv - eu$. Then

$$D\,z = df + ude + (u-1)e^2. \tag{4}$$

So D is a *positive* integer, even though dv and eu largely cancel, and it is small when the drops are small. Also $(q, x) = (\lfloor a/b_i \rfloor, b_i)$ lies on the curve $xM(q) = aN(q) + D$, where M and N are explicit polynomials depending on (u, v) (Lemma 3.3). A resultant-type identity shows that each such curve has $a^{o(1)}$ admissible points (Lemma 4.1). The number of triples (u, v, D) is $\ll (B/T)^2 \cdot a/T^3$, where $B = a/A$, which gives (3).

A second positive integer. Six consecutive terms carry three overlapping curvatures D, E, F and four quotient increments u, v, w, z. In Section 7 we show that the integer

$$L = \frac{S(v,w,z)\,D - S(u,v,w)\,E}{v}, \qquad S(u,v,w) = u^2v + (u-1)v^2 + uw,$$

satisfies the exact identity $qwL = J(u,v,w,z)\,E + S(u,v,w)\,F$, where J is a polynomial with positive values (Lemma 7.1). Hence $L \ge 1$, and $q \asymp a/A$ is bounded by a polynomial in the small parameters u, v, w, z, E, F. A single window of six terms with small total drop then gives

$$T^6 A^2 \le 96 \cdot 10^6\, a^3 \qquad (T \ge 10), \tag{5}$$

with no divisor function at all (Proposition 7.3).

With (5) in hand, the only remaining loss is in the drop bound (2). That loss comes from the number of divisors of $a+h$ in $[A, 2A)$ as h ranges over a short interval. Section 8 shows that any bound $\ll y\,G(x)$ for such counts over intervals $(x, x+y]$ with $y \geq x^{1/4}$, uniform in A and with $G(x) \ll x^{1/40}$, gives $P(a,b) \ll a^{3/10}G(a)^{1/5}$ (Theorem 8.4). The counts are majorised by the Erdős–Hooley function. Combining the short-sum theorem of Nair and Tenenbaum [9] with the recent mean-value bound of de la Bretèche and Tenenbaum [2] gives $G(x) = (\log\log x)^{5/2}$, and hence the following.

Theorem 1.3. *Assume the theorem of Nair–Tenenbaum stated as* (9) *and the theorem of de la Bretèche–Tenenbaum stated as* (10) *below. Then*

$$P(a,b) \ll a^{3/10}\sqrt{\log\log 3a} \qquad \textit{uniformly for integers } 0 < b < a.$$

Both hypotheses are published theorems. They are hypotheses only in the sense of the formalisation: we have not formalised their proofs. Any other short-interval divisor bound can be used in their place. For example, Shiu's theorem [15] gives $\sum_{x<n\leq x+y}\tau(n) \ll y\log x$ for $y \geq x^{1/4}$, and hence $P(a,b) \ll a^{3/10}(\log a)^{1/5}$; we have not formalised that either.

Structured trajectories and the Hankel hierarchy. Write $k_i = q_{i+1} - q_i$ for the quotient increments, and put

$$R = 1 + \#\{i : 1 \leq i,\ i+2 \leq P(a,b),\ k_i < k_{i+1}\}.$$

Thus $R-1$ counts the rises between consecutive increments.

Theorem 1.4. $P(a,b) \ll a^{1/4}\sqrt{R}$ *uniformly for* $0 < b < a$. *In particular, if the quotient increments never rise, then* $P(a,b) \ll a^{1/4}$.

The proof (Section 9) combines three facts. First, $D \geq 1$ says that d_i/k_i is strictly decreasing, so the drops strictly decrease along every stretch without a rise. Second, the six-step integer L gives an exact identity for six quotients in arithmetic progression, which forces their common difference to be large (Lemma 9.1). Third, a counting lemma for sequences in which no value occurs too often. So an orbit of length about $a^{3/10}$ must contain $\gg a^{1/10}$ rises.

Finally, both D and L fit into a single structure (Section 10). The orbit carries an exact finite model by positive measures ν_i with $\int d\nu_i = b_i/a$. The Hankel determinants $I_r(i) = a\det\big(\int x^{j+l}\,d\nu_i\big)_{j,l<r}$ are positive integers whenever the remaining orbit has at least r positive terms. The first four agree with b_i, $b_i - b_{i+1}$, D and L in the nonterminal ranges specified in Theorem 10.2.

Below 3/10**.** The following four elementary bounds explain the pattern-counting ingredients and an alternative lattice refinement.

Theorem 1.5. *For every* $\varepsilon > 0$ *there is a constant* C_ε *such that*

$$P(a,b) \leq C_\varepsilon\, a^{23/77+\varepsilon} \qquad \textit{for all integers } 0 < b < a.$$

Here $23/77 = 0.298701\ldots$, a saving of $1/770$ over $3/10$. The proof (Section 11) uses the fourth Hankel invariant $I_4 = L$ as an integer, not only as a positive number. At the critical configuration L has bounded size. The labels $L(i), L(i+1), L(i+2)$ and four cofactors of an explicit 3×4 integer matrix form a *pattern* of seven integers. Two consecutive states give two linear equations and one quadratic equation in five consecutive quotient increments, and the quadratic has a positive leading coefficient. An exact identity $I_4(i)\,\pi_i(-q_i) = a\det(\mu_i(x^{j+l}))_{j,l<4}$, with π_i the degree-four orthogonal polynomial of the moment model, makes q_i a root of a quartic that is fixed by the pattern and the increments. So each pattern accounts for $O(H^2)$

states, where H bounds the increments. The number of patterns is controlled by Hadamard's inequality applied to the moment model.

Counting the solutions of the quadratic equation more carefully gives a further improvement.

Theorem 1.6. *For every $\varepsilon > 0$ there is a constant C_ε such that*

$$P(a,b) \le C_\varepsilon\, a^{11/37+\varepsilon} \qquad \text{for all integers } 0 < b < a.$$

Here $11/37 = 0.297297\ldots$, a saving of $1/370$ over $3/10$. The extra input (Section 12) is that the quadratic part is nondegenerate with an indefinite binary restriction, by exact sign identities for the cofactors. So each slice is a binary norm equation, which has $a^{o(1)}$ solutions of size $a^{O(1)}$, uniformly in its coefficients, when its right-hand side is nonzero. There are at most two exceptional slices, which are counted separately.

Theorem 1.7. *For every $\varepsilon > 0$ there is a constant C_ε such that*

$$P(a,b) \le C_\varepsilon\, a^{8/27+\varepsilon} \qquad \text{for all integers } 0 < b < a.$$

Here $8/27 = 0.296296\ldots$. The seven pattern integers are signed coordinates of the vector of 3×3 minors of a 3×5 integer matrix (Section 13). All ten minors are bounded by X, and such minor vectors number $X^{5+o(1)}$ rather than X^7.

Theorem 1.8. *For every $\varepsilon > 0$ there is a constant C_ε such that*

$$P(a,b) \le C_\varepsilon\, a^{45/152+\varepsilon} \qquad \text{for all integers } 0 < b < a.$$

Here $45/152 = 0.296052\ldots$. The pivot gaps of a state lie in a lattice $\Lambda_p \subseteq \mathbb{Z}^3$ of index at most X determined by its minor vector p (Section 14). If the shortest vector of Λ_p has length λ, a reduced basis cuts the ternary quadratic into $O(H/\lambda+1)$ slices, and only $X^{4+o(1)}t^3$ minor vectors have $\lambda \le t$. This gives $T \ll a^{o(1)}(HX^{14/3} + X^5)$ in place of $T \ll a^{o(1)}HX^5$.

The joint-invariant argument returns to the ordinary minor-pattern count of Section 13 and improves the count within each pattern. Its additional input is the affine-curve theorem used in Section 15.

Formal verification. Every step of the proofs of Theorems 1.1, 1.2, 1.5, 1.6, 1.7 and 1.8 has been formalised in Lean 4 [7] using Mathlib [8]. So have Proposition 7.3, Theorem 8.4, the deduction of Theorem 1.3 from (9) and (10), Theorem 1.4 and Theorem 10.2. Section 16 describes the formalisation. The constants in this paper are the constants in the Lean proof. The accompanying repository contains the proof source, independent submission statements and build instructions.

What is not proved. These results are far from (1), and the method cannot reach it; see Section 17.

2. Preliminaries

2.1. **Notation.** Throughout, $a > b > 0$ are integers, $b_1 = b$, $b_{i+1} = a \bmod b_i$, and $P = P(a,b)$, so that $a > b_1 > b_2 > \cdots > b_P > 0 = b_{P+1}$. Write $q_i = \lfloor a/b_i \rfloor$, so

$$a = q_i b_i + b_{i+1}, \qquad 0 \le b_{i+1} < b_i \qquad (1 \le i \le P).$$

Let $\tau(n)$ be the number of positive divisors of n, and let

$$\Delta = \Delta(a) = \max_{1 \le n \le 32a^7} \tau(n).$$

We write $F \ll_\eta G$ when $F \le C_\eta G$, with the constant uniform in the dividend and orbit. A factor denoted $a^{o(1)}$ is bounded by $C_\eta a^\eta$ for every fixed $\eta > 0$; such factors include the divisor losses below. We write $F \asymp G$ when both $F \ll G$ and $G \ll F$ hold.

Lemma 2.1 (Quotients increase)**.** *If $b_{i+1} > 0$ then $q_{i+1} \geq q_i + 1$.*

Proof. Since $b_{i+1} < b_i$ we have $(q_i + 1)b_{i+1} \leq q_i b_i + b_{i+1} = a$, so $q_{i+1} = \lfloor a/b_{i+1} \rfloor \geq q_i + 1$. □

Lemma 2.2 (Divisor bound)**.** *For every integer $k \geq 1$ there is C_k with $\tau(n)^k \leq C_k\, n$ for all $n \geq 1$.*

Proof. This is standard: $\tau(n)^k/n = \prod_{p^\alpha \| n} (\alpha+1)^k/p^\alpha$. Each factor with $p \geq 2^k$ is at most $(\alpha+1)^k/2^{k\alpha} \leq 1$. Each of the fewer than 2^k factors with $p < 2^k$ is at most $\sup_\alpha (\alpha+1)^k/2^\alpha < \infty$. □

2.2. **Window runs.**

Definition 2.3. Let $A \geq 1$ be an integer. A *window run* of length T is a sequence $x_0, x_1, \ldots, x_{T-1}$ of consecutive terms of the sequence (b_i), so $x_{j+1} = a \bmod x_j$, all lying in $[A, 2A)$.

In a window run, $x_0 > x_1 > \cdots > x_{T-1}$ and $x_j \leq a$.

3. The curvature of three consecutive steps

Fix four consecutive terms

$$x = b_i, \quad y = b_{i+1} = a \bmod x, \quad z = b_{i+2} = a \bmod y, \quad w = b_{i+3} = a \bmod z,$$

with $x, y, z > 0$ (and $w \geq 0$). Write

$$q = \left\lfloor \frac{a}{x} \right\rfloor, \quad u = \left\lfloor \frac{a}{y} \right\rfloor - q, \quad v = \left\lfloor \frac{a}{z} \right\rfloor - \left\lfloor \frac{a}{y} \right\rfloor, \qquad d = x - y, \quad e = y - z, \quad f = z - w.$$

By Lemma 2.1, $u, v \geq 1$, and clearly $d, e, f \geq 1$. The division equations read

$$xq + y = a, \qquad y(q+u) + z = a, \qquad z(q+u+v) + w = a. \tag{6}$$

Definition 3.1. The *curvature* of the block is $D = dv - eu$.

Lemma 3.2 (Curvature identity)**.**

$$D\,z = df + ude + (u-1)e^2.$$

In particular $D \geq 1$.

Proof. Subtracting the first two equations of (6) gives $uy = qd + e$, hence $uz = uy - ue = qd + (1-u)e$. Subtracting the last two gives $vz = (q+u)e + f$. Therefore

$$(dv - eu)z = d\big((q+u)e + f\big) - e\big(qd + (1-u)e\big) = df + ude + (u-1)e^2.$$

The right-hand side is at least $df \geq 1$, and $z > 0$, so $D > 0$. □

Lemma 3.3 (The fiber equation)**.** *Put $M(q) = uq^2 + (u^2+u+v)q + v$ and $N(q) = uq + u^2 + v$. Then*

$$x\,M(q) = a\,N(q) + D.$$

Proof. From (6), $d = (q+1)x - a$ and $e = y - z = (q+u+1)y - a = a(q+u) - q(q+u+1)x$. Substituting these into $D = dv - eu$ gives $D = x\big(uq(q+u+1) + v(q+1)\big) - a\big(u(q+u) + v\big)$, which is the claim. □

Lemma 3.4 (Size of the parameters)**.** *If $x \leq a$ then*

$$u\,yx \leq a(d+e), \qquad v\,zy \leq a(e+f), \qquad D\,z \leq (u+1)(d+e+f)^2.$$

Proof. From $uy = qd + e$ we get $uyx = (qx)d + ex \leq ad + ae$, since $qx \leq a$ and $x \leq a$. Similarly $vz = (q+u)e + f$ with $(q+u)y \leq a$ and $y \leq a$ gives the second bound. For the third, by Lemma 3.2, $Dz = df + ude + (u-1)e^2 \leq df + u\,e(d+e) \leq (1+u)(d+e+f)^2$. □

4. Few blocks share a triple (u, v, D)

Lemma 4.1 (Fiber count)**.** *Let $a \geq 1$, $1 \leq u, v \leq a$ and $1 \leq D \leq 2a$. Let $M(q) = uq^2 + pq + v$ with $p = u^2 + u + v$, and let $\Lambda(q) = cq + \ell$ with $c = au$ and $\ell = a(u^2 + v) + D$. Then the set*

$$Q = \{q \in \mathbb{Z}_{\geq 0} : \ xM(q) = \Lambda(q) \text{ for some integer } x \geq 1\}$$

has at most $\Delta = \max_{1 \leq n \leq 32a^7} \tau(n)$ elements.

Proof. Put $U(q) = ucq + cp - u\ell$ and $R = u\ell^2 - cp\ell + c^2 v$. Expanding shows

$$c^2 M(q) - U(q)\Lambda(q) = R \tag{7}$$

identically in q. If $q \in Q$ with witness x, substituting $\Lambda(q) = xM(q)$ into (7) gives

$$R = M(q)\big(c^2 - U(q)x\big). \tag{8}$$

We bound the sizes first. From $u, v \leq a$ and $D \leq 2a$ we get $\ell \leq 4a^3$, $p \leq 3a^2$ and $c \leq a^2$, so

$$|R| \leq u\ell^2 + cp\ell + c^2 v \leq 16a^7 + 12a^7 + a^5 \leq 32a^7, \qquad c^2 \leq a^4 \leq 32a^7.$$

Case $R \neq 0$. By (8), $M(q)$ is a positive divisor of $|R|$. Since $u, p \geq 1$, the map $q \mapsto M(q)$ is strictly increasing on $q \geq 0$, hence injective. Therefore $|Q| \leq \tau(|R|) \leq \Delta$.

Case $R = 0$. Since $M(q) > 0$, (8) gives $U(q)x = c^2$. Here $c^2 > 0$ and $x \geq 1$, so $U(q)$ is a positive divisor of c^2. Since U is linear with positive leading coefficient uc, it is injective. Therefore $|Q| \leq \tau(c^2) \leq \Delta$. □

The case $R = 0$ cannot be dropped: it is the case in which M and Λ might share a factor.

5. Counting visits to a window

In this section $x_0, \ldots, x_{T-1}$ is a window run in $[A, 2A)$, and $q_j = \lfloor a/x_j \rfloor$ are its quotients. We continue to write $\Delta = \max_{1 \leq n \leq 32a^7} \tau(n)$.

Proposition 5.1 (Drop bound; after Erdős–Shallit)**.** *If $T \geq 2$ then $T(T-1) \leq 4A\Delta$.*

Proof. Let $h_j = x_j - x_{j+1} \geq 1$ for $0 \leq j \leq T-2$. By telescoping, $\sum_j h_j = x_0 - x_{T-1} \leq (2A-1) - A = A - 1$. Call j *good* if $h_j(T-1) \leq 2A$ and *bad* otherwise. Each bad j has $h_j(T-1) \geq 2A+1$, so

$$|\,\mathrm{Bad}\,|\,(2A+1) \leq (T-1)\textstyle\sum_j h_j \leq (T-1)(A-1).$$

If $2|\,\mathrm{Bad}\,| \geq T-1$ this would give $(T-1)(2A+1) \leq 2(T-1)(A-1)$, which is false. Hence $2|\,\mathrm{Bad}\,| < T-1$, and since $|\,\mathrm{Good}\,| + |\,\mathrm{Bad}\,| = T-1$ we get $2|\,\mathrm{Good}\,| \geq T$.

For good j we have $1 \leq h_j \leq H := \lfloor 2A/(T-1) \rfloor$. Also $a + h_j = q_j x_j + x_{j+1} + x_j - x_{j+1} = (q_j + 1)x_j$, so $x_j \mid a + h_j$. Moreover $a + h_j \leq 2a \leq 32a^7$. Since $j \mapsto x_j$ is injective, at most $\tau(a + h) \leq \Delta$ good indices have $h_j = h$. Hence $|\,\mathrm{Good}\,| \leq H\Delta$, and

$$T(T-1) \leq 2|\,\mathrm{Good}\,|(T-1) \leq 2\Delta\, H(T-1) \leq 4A\Delta.$$ □

Proposition 5.2 (Curvature bound)**.** *If $T \geq 12$ then $T^6 A^2 \leq 995328\, a^3 \Delta$.*

Proof. For $0 \leq j \leq T-4$ let $s_j = x_j - x_{j+3}$ be the total drop of the block $x_j, x_{j+1}, x_{j+2}, x_{j+3}$. By telescoping,

$$\sum_{j=0}^{T-4} s_j = (x_0 + x_1 + x_2) - (x_{T-3} + x_{T-2} + x_{T-1}) \leq 3(2A-1) - 3A = 3A - 3.$$

Call j *good* if $s_jT \le 12A$ and *bad* otherwise. Then $|\,\mathrm{Bad}\,|(12A+1) \le T(3A-3)$. If $4|\,\mathrm{Bad}\,| \ge T$ this would give $T(12A+1) \le 4T(3A-3)$, which is false. So $4|\,\mathrm{Bad}\,| \le T-1$. Since $|\,\mathrm{Good}\,| = T-3-|\,\mathrm{Bad}\,|$ and $T \ge 12$,

$$2|\,\mathrm{Good}\,| \ge 2T-6-\tfrac{T-1}{2} \ge T.$$

Fix a good j. Apply Section 3 to $x = x_j$, so that $y, z, w = x_{j+1}, x_{j+2}, x_{j+3}$, all in $[A, 2A)$. Write $s = s_j = d+e+f$. Lemma 3.4 gives $uA^2 \le uyx \le a(d+e) \le as$. Multiplying by T and using $sT \le 12A$ gives $uA^2T \le 12aA$, that is,

$$u\,AT \le 12a, \qquad \text{and likewise} \qquad v\,AT \le 12a.$$

Also $DA \le Dz \le (u+1)s^2 \le 2us^2$. Multiplying by T^2 gives $DAT^2 \le 2u(sT)^2 \le 288uA^2$, so $DT^2 \le 288uA$, and then

$$D\,T^3 \le 288\,u\,AT \le 3456\,a.$$

Thus the triple (u, v, D) of a good block lies in $[1, U]^2 \times [1, D^*]$, where

$$U = \lfloor 12a/(AT) \rfloor \le a, \qquad D^* = \lfloor 3456a/T^3 \rfloor \le 2a.$$

The inequalities $U \le a$ and $D^* \le 2a$ hold because $AT \ge 12$ and $T^3 \ge 1728$.

Now fix such a triple and consider the good blocks that have it. By Lemma 3.3, each one gives a point $q = q_j = \lfloor a/x_j \rfloor$ of the set Q of Lemma 4.1. The map $j \mapsto q_j$ is injective on these blocks: q_j determines $x_j = L(q_j)/M(q_j)$, and x_j determines j. By Lemma 4.1, at most Δ good blocks share a triple. So $|\,\mathrm{Good}\,| \le U^2D^*\Delta$, and

$$T^6A^2 = T\,(AT)^2\,T^3 \le 2\Delta\,(U\,AT)^2\,(D^*T^3) \le 2\Delta(12a)^2(3456a) = 995328\,a^3\Delta. \qquad \square$$

Corollary 5.3 (Window bound)**.** *If $T \ge 12$ then $T^{10} \le 63700992\,a^3\Delta^3$.*

Proof. Using $T \le 2(T-1)$ and the two propositions,

$$T^{10}A^2 \le 4\,(T^6A^2)\,\big(T(T-1)\big)^2 \le 4 \cdot 995328\,a^3\Delta \cdot (4A\Delta)^2 = 63700992\,a^3\Delta^3A^2.$$

Divide by A^2. $\square$

Remark 5.4. In terms of $B = a/A$, Propositions 5.1 and 5.2 say $T \ll a^{1/2+o(1)}B^{-1/2}$ and $T \ll a^{1/6+o(1)}B^{1/3}$. The first is better when $B > a^{2/5}$, and both equal $a^{3/10+o(1)}$ at $B = a^{2/5}$. Corollary 5.3 takes the product of the two bounds, so no case split is needed.

6. Proof of Theorem 1.2

Let $0 < b < a$. For $0 \le k \le \log_2 a$ let T_k be the number of indices $i \le P$ with $2^k \le b_i < 2^{k+1}$. Every b_i with $i \le P$ lies in $[1, a)$, so $P = \sum_{0 \le k \le \log_2 a} T_k$. Since (b_i) is decreasing, the indices counted by T_k are consecutive. So when $T_k > 0$ they form a window run in $[A, 2A)$ with $A = 2^k$.

Let $\varepsilon > 0$. Choose an integer $m > 7/(5\varepsilon)$, and let C_{3m} be the constant of Lemma 2.2 with $k = 3m$. The maximum Δ is attained at some $n_0 \le 32a^7$, so $\Delta^{3m} \le C_{3m} \cdot 32a^7$. If $T_k \ge 12$, Corollary 5.3 gives

$$T_k^{10m} \le 63700992^m\,a^{3m}\,\Delta^{3m} \le K\,a^{3m+7}, \qquad K = 63700992^m \cdot C_{3m} \cdot 32,$$

and hence

$$T_k \le K^{1/(10m)}\,a^{3/10+7/(10m)} \le K^{1/(10m)}\,a^{3/10+\varepsilon/2}.$$

If $T_k \le 11$ this bound holds with 11 in place of $K^{1/(10m)}$. Finally $\lfloor \log_2 a \rfloor + 1 \le C'_\varepsilon a^{\varepsilon/2}$. Summing over k,

$$P(a,b) \le (\lfloor \log_2 a \rfloor + 1)\big(11 + K^{1/(10m)}\big)a^{3/10+\varepsilon/2} \le C_\varepsilon\,a^{3/10+\varepsilon}. \qquad \square$$

7. Six consecutive steps

Fix six consecutive terms $x_0 > x_1 > \cdots > x_5 \geq 0$ of the sequence (b_i) with $x_4 > 0$, and write their quotients as

$$\lfloor a/x_0 \rfloor = q, \quad \lfloor a/x_1 \rfloor = q+u, \quad \lfloor a/x_2 \rfloor = q+u+v,$$
$$\lfloor a/x_3 \rfloor = q+u+v+w, \quad \lfloor a/x_4 \rfloor = q+u+v+w+z.$$

(In this section w and z are quotient increments, not terms of the sequence.) By Lemma 2.1, $u, v, w, z \geq 1$. Let $f_k = x_k - x_{k+1} \geq 1$ be the drops, and let

$$D = f_0 v - f_1 u, \qquad E = f_1 w - f_2 v, \qquad F = f_2 z - f_3 w$$

be the curvatures of the blocks starting at x_0, x_1 and x_2. Write $\mathcal{E}_k$ for the division equation $x_k q_k + x_{k+1} - a = 0$, where $q_k = \lfloor a/x_k \rfloor$. Put

$$S(u,v,w) = u^2 v + (u-1)v^2 + uw,$$
$$J(u,v,w,z) = \big(u^2 + (u-1)v\big)z + w\big(u^3 + uv(2u-1) + (u-1)v^2\big) + w^2\big((u-1)(u+v)+1\big),$$
$$L = (vw + w^2 + z)\, D - \big(u^2 + (u-1)v\big)E - w\,(wf_0 - uf_2).$$

Lemma 7.1 (The six-step identity)**.** (1) $S(v,w,z)\, D - S(u,v,w)\, E = v\, L$.
(2) $q\, w\, L = J(u,v,w,z)\, E + S(u,v,w)\, F$.

Proof. Part (1) is a polynomial identity: since $S(v,w,z) \equiv -w^2$ and $S(u,v,w) \equiv uw \pmod v$, and $wD + uE = v(wf_0 - uf_2)$, the difference is divisible by v, and expanding gives L. Part (2) holds modulo the division equations. Explicitly, $qwL - JE - S(u,v,w)F$ equals

$$w(v^2 w + vw^2 + vz - w^2)\,\mathcal{E}_0 - w(u^2 w + 2uvw + uw^2 + uz + v^2 w + vw^2 - vw + vz - w^2)\,\mathcal{E}_1$$
$$+ w(u^2 v + u^2 w + uv^2 + 2uvw + uw^2 + uw + uz - v^2 - vw)\,\mathcal{E}_2 - w(u^2 v + uv^2 + uw - v^2)\,\mathcal{E}_3.$$

This is an identity of polynomials in a, q, the x_k and the increments, checked by expansion. □

Corollary 7.2. *Suppose $q \geq 1$ and $u, v, w, z \leq m$. Then $E, F \geq 1$, $L \geq 1$, and*

$$q \leq 9m^3 E + 3m^3 F.$$

Proof. Lemma 3.2, applied to the blocks starting at x_1 and x_2, gives $E, F \geq 1$. Clearly $J > 0$ and $S \geq 0$, so by Lemma 7.1(2) $qwL > 0$. Since L is an integer, $L \geq 1$ and $qw \leq JE + SF$. Bounding each monomial gives $J \leq 9m^3 w$ and $S(u,v,w) \leq 3m^3 w$. Dividing by w gives the claim. □

So the quotient q, which is about a/A, is controlled by the small parameters of a single block of six terms. No counting over many blocks is needed.

Proposition 7.3 (Six-step window bound)**.** *For a window run of length $T \geq 10$,*

$$T^6 A^2 \leq 96 \cdot 10^6\, a^3.$$

Proof. For $0 \leq j \leq T-6$ let $s_j = x_j - x_{j+5}$. By telescoping, $\sum_{j \leq T-6} s_j \leq 5(2A-1) - 5A = 5A-5$. So some j has $s_j(T-5) \leq 5A-5$, and since $T \leq 2(T-5)$, we get $s := s_j$ with $sT \leq 10A$. Apply Corollary 7.2 to $x_j, \ldots, x_{j+5}$, all of which lie in $[A, 2A)$, with $m = \max(u,v,w,z)$.

By Lemma 3.4 at x_j, x_{j+1} and x_{j+2}, each increment satisfies (increment) $\cdot\, A^2 \leq as$. Multiplying by T gives $mAT \leq 10a$. By Lemma 3.4 again, $EA \leq Ex_{j+3} \leq (v+1)s^2 \leq 2ms^2$, and likewise $FA \leq 2ms^2$. Hence $qA \leq 24m^4 s^2$. Also $a = qx_j + x_{j+1} < (q+1)x_j < 2A(q+1) \leq 4Aq$. Therefore

$$qA \cdot A^4 T^6 \leq 24\,(mAT)^4 (sT)^2 \leq 24 \cdot 10^6\, a^4 A^2,$$

so $qA^3 T^6 \leq 24 \cdot 10^6 a^4$, and $aA^2 T^6 \leq 4qA^3 T^6 \leq 96 \cdot 10^6\, a^4$. □

Remark 7.4. In terms of $B = a/A$, Proposition 7.3 reads $T \ll a^{1/6}B^{1/3}$. This is Proposition 5.2 without the factor $\Delta^{1/6}$. Combined with Proposition 5.1 as in Corollary 5.3, it gives $T^{10} \ll a^3\Delta^2$ and another proof of Theorem 1.2. The Lean proof of Theorem 1.2 uses Corollary 5.3.

8. Divisors over short intervals

For $A \geq 1$ let $\tau_A(n) = \#\{d \mid n : \ A \leq d < 2A\}$.

Proposition 8.1 (Drop count)**.** *For a window run of length $T \geq 2$ and any $Y \geq \lfloor 2A/(T-1) \rfloor$,*

$$T \leq 2 \sum_{a<n\leq a+Y} \tau_A(n).$$

Proof. This is the proof of Proposition 5.1, stopped before the divisor bound. The good indices j have $1 \leq h_j \leq \lfloor 2A/(T-1) \rfloor$, and $j \mapsto x_j$ maps those with $h_j = h$ injectively into the divisors of $a + h$ in $[A, 2A)$. □

Definition 8.2. Let $G\colon \mathbb{N} \to \mathbb{R}$. We say that $\mathrm{SD}(G)$ holds if there are C, x_0 such that for all integers $x \geq x_0$, $x^{1/4} \leq y \leq x$ and $A \geq 1$,

$$\sum_{x<n\leq x+y} \tau_A(n) \leq C\, y\, G(x).$$

Lemma 8.3 (Geometric sums)**.** *Let $\alpha, \beta, s, t > 0$ and $K \geq 0$. Then*

$$\sum_{k<K} \min\bigl(\alpha 2^{-sk},\ \beta 2^{tk}\bigr) \leq \Bigl(\frac{1}{1-2^{-s}} + \frac{2^t}{2^t-1}\Bigr)\alpha^{t/(s+t)}\beta^{s/(s+t)}.$$

Proof. Let $M = \alpha^{t/(s+t)}\beta^{s/(s+t)}$. Then $(\alpha 2^{-sk})^t(\beta 2^{tk})^s = M^{s+t}$, so for each k either $\beta 2^{tk} \leq M$ or $\alpha 2^{-sk} < M$. The terms $\beta 2^{tk} \leq M$ grow with ratio 2^t, so they sum to at most $M2^t/(2^t-1)$. The terms $\alpha 2^{-sk} < M$ decay with ratio 2^{-s}, so they sum to at most $M/(1-2^{-s})$. □

Theorem 8.4. *Suppose $\mathrm{SD}(G)$ holds, $G \geq 1$, and $G(x) \ll x^{1/40}$. Then $P(a,b) \ll a^{3/10}\, G(a)^{1/5}$ uniformly for $0 < b < a$.*

Proof. Small a are handled by $P(a,b) \leq b < a$. Fix a dyadic window $[A, 2A)$, $A = 2^k$, with $T \geq 20$ visits. Proposition 7.3 gives $T \leq \alpha A^{-1/3}$, where $\alpha = (96 \cdot 10^6)^{1/6}a^{1/2}$. Take $Y = \max(\lfloor 2A/(T-1) \rfloor, \lceil a^{1/4} \rceil) \leq a$ in Proposition 8.1. With $H = \lfloor 2A/(T-1) \rfloor$, $\mathrm{SD}(G)$ gives $T \leq 2CG(a)(H + a^{1/4} + 1)$. Multiplying by $T-1$, using $H(T-1) \leq 2A$, and solving the quadratic gives $T \leq 1 + 2CG(a)(a^{1/4}+1) + 2\sqrt{CG(a)A}$, hence, as $T \geq 20$, $T \leq \gamma + \beta A^{1/2}$, where $\beta = 4\sqrt{CG(a)}$ and $\gamma = 4CG(a)(a^{1/4}+1)$. So every window satisfies

$$T_k \leq 19 + \gamma + \min\bigl(\alpha\, 2^{-k/3},\ \beta\, 2^{k/2}\bigr).$$

Lemma 8.3 with $s = 1/3$ and $t = 1/2$, where $1/(1-2^{-1/3}) + \sqrt{2}/(\sqrt{2}-1) < 9$, gives

$$\sum_k \min\bigl(\alpha\, 2^{-k/3},\ \beta\, 2^{k/2}\bigr) \leq 9\, \alpha^{3/5}\beta^{2/5} \ll a^{3/10}G(a)^{1/5},$$

with no factor $\log a$. The remaining terms contribute $(\log_2 a + 1)(19 + \gamma) \ll a^{1/100} \cdot a^{1/4}G(a)$. This is $\ll a^{3/10}G(a)^{1/5}$, because $G(a)^{4/5} \ll a^{1/50}$. □

The Erdős–Hooley function. Let $\Delta_{\mathrm{H}}(n) = \max_{u\in\mathbb{R}} \#\{d \mid n : \ e^u < d \le e^{u+1}\}$. Taking $e^{u+1} = 2A$, and using $e > 2$, gives $\tau_A(n) \le \Delta_{\mathrm{H}}(n)$. We use two published theorems.

(NT) Nair and Tenenbaum [9] prove a general upper bound for short sums of functions F satisfying $F(mn) \le \min(\lambda_1^{\Omega(m)}, \lambda_2 m^{\varepsilon})F(n)$ for $(m,n) = 1$. Since $\Delta_{\mathrm{H}}(mn) \le \tau(m)\Delta_{\mathrm{H}}(n)$ for $(m,n) = 1$, the function Δ_{H} satisfies this with $\lambda_1 = 2$ for every $\varepsilon > 0$. Their bound, for the polynomial $Q(X) = X$, gives for every fixed $\sigma > 0$, x large and $x^\sigma \le y \le x$

$$\sum_{x<n\le x+y} \Delta_{\mathrm{H}}(n) \ll_\sigma \frac{y}{\log x} \sum_{n\le x} \frac{\Delta_{\mathrm{H}}(n)}{n}. \tag{9}$$

We use it with $\sigma = 1/4$. In the parameters of their Theorem 1, the polynomial degree is $g = \deg Q = 1$; take $\varepsilon = 1/16$ and $\delta = 1$. Then $4g^2\varepsilon = 1/4$ and the required class is $\mathcal{M}_1(2, B, 1/48)$ for a suitable B. The same consequence is displayed immediately after equation (2) of their paper.

(BT) de la Bretèche and Tenenbaum [2, Theorem 1.1] prove that for large x

$$\sum_{n\le x} \Delta_{\mathrm{H}}(n) \ll x\,(\log\log x)^{5/2}. \tag{10}$$

Lemma 8.5. *Assume* (10)*, and let* $\ell(x) = \max(1, \log\log x)$*. Then* $\sum_{n\le x} \Delta_{\mathrm{H}}(n)/n \ll \log x \cdot \ell(x)^{5/2}$ *for large* x*.*

Proof. Split n into dyadic blocks $2^j \le n < 2^{j+1} \le 2x$. The terms with n below the threshold of (10) contribute $O(1)$, since $\Delta_{\mathrm{H}}(n) \le n$. Each remaining block contributes at most $2^{-j}\sum_{n\le 2^{j+1}} \Delta_{\mathrm{H}}(n) \ll \ell(2x)^{5/2}$, and $\ell(2x) \le 2\ell(x)$. There are at most $\log_2 x + 1 \le 3\log x$ blocks. □

Proof of Theorem 1.3. By (9) and Lemma 8.5, $\sum_{x<n\le x+y} \tau_A(n) \le \sum_{x<n\le x+y} \Delta_{\mathrm{H}}(n) \ll y\,\ell(x)^{5/2}$ for $x^{1/4} \le y \le x$. So SD(G) holds with $G = \ell^{5/2}$, which satisfies $G \ge 1$ and $G(x) \ll x^{1/40}$. Theorem 8.4 gives $P(a,b) \ll a^{3/10}\ell(a)^{1/2} \ll a^{3/10}\sqrt{\log\log 3a}$. □

9. Rises in the quotient increments

Throughout this section $k_i = q_{i+1} - q_i \ge 1$, $d_i = b_i - b_{i+1} \ge 1$, and a *rise* is an index $1 \le i$ with $i + 2 \le P(a,b)$ and $k_i < k_{i+1}$.

Lemma 9.1 (Six quotients in arithmetic progression)**.** *Let* $q, h \ge 1$ *and* $x_0 > x_1 > \cdots > x_6 \ge 0$ *satisfy* $x_j(q + jh) + x_{j+1} = a$ *for* $0 \le j \le 5$*. Then* $q^4 \le 324\,h^3 a$*.*

Proof. Let $f_j = x_j - x_{j+1}$, and let $E = (f_1 - f_2)h$ and $F = (f_2 - f_3)h$ be the curvatures of the blocks starting at x_1 and x_2 (all their increments equal h). With $U = f_0 - 2f_1 + f_2$, the division equations $\mathcal{E}_j\colon x_j(q + jh) + x_{j+1} - a = 0$ give the exact identity

$$qhU = (3h-1)E + F, \qquad \text{namely} \qquad qhU - (3h-1)E - F = h(\mathcal{E}_0 - 3\mathcal{E}_1 + 3\mathcal{E}_2 - \mathcal{E}_3).$$

This is the case $u = v = w = z = h$ of Lemma 7.1, where $L = 2h^3U$. By Lemma 3.2, $E, F \ge 1$, so the integer U is at least 1.

Now bound E and F. Subtracting consecutive division equations gives $q_j f_j \le h x_{j+1}$, where $q_j = q + jh$. Since $x_{j+1}q_{j+1} \le a$, this yields $q^2 f_j \le ha$ for $1 \le j \le 4$. By Lemma 3.4, $Ex_3 \le (h+1)(f_1+f_2+f_3)^2 \le 9h^2(h+1)a^2/q^4$. Also $a < x_3(q+3h+1)$, so $Eq^4 \le 9h^2(h+1)(q+3h+1)a$, and likewise $Fq^4 \le 9h^2(h+1)(q+4h+1)a$. Hence

$$q^5 h \le q^4\big((3h-1)E + F\big) \le 27h^3(h+1)(q+4h+1)a.$$

If $q \ge h$ this gives $q^4 \le 324h^3a$. If $q < h$ then $q^4 < h^4 \le h^3 a$, since $h \le q_1 \le a$. □

Remark 9.2. An earlier version of this argument proved Lemma 9.1 for seven quotients by an analytic argument: it represents a periodic continuation of the orbit by integrals of positive functions, and compares finite differences. The identity above gives an algebraic proof with one fewer quotient.

Lemma 9.3. *Whenever $b_{i+2} > 0$, $d_{i+1}k_i < d_i k_{i+1}$. In particular, if $k_{i+1} \le k_i$ then $d_{i+1} < d_i$.*

Proof. This is $D = d_i k_{i+1} - d_{i+1}k_i \ge 1$ (Lemma 3.2). If $k_{i+1} \le k_i$ and $d_{i+1} \ge d_i$, then $d_i k_{i+1} \le d_{i+1}k_i$, which is a contradiction. □

Lemma 9.4. *Let $c_j \ge 1$ be integers, $j \in J$, such that no value occurs more than M times. Then $|J|^2 \le 2M \sum_{j\in J} c_j$.*

Proof. Induct on $|J|$. Let c_{j_0} be the largest value. The values lie in $[1, c_{j_0}]$, so $|J| \le M c_{j_0}$. Removing j_0 and using $|J|^2 = (|J|-1)^2 + 2|J| - 1$ completes the induction. □

Proposition 9.5. *Let $x_0, \dots, x_{T-1}$ be a window run in $[A, 2A)$ with $T \ge 2$, and let ρ be the number of rises among its $T-1$ increments. Then:*

(1) $(T-1)^2 \le 2(\rho+1)(A-1)$;
(2) $T - 1 \le \lfloor a/A \rfloor$;
(3) *if $6A \le a$, then $T - 1 = \ell + s$ with $\ell^2 \le 8(\rho+1)\lfloor a/A \rfloor$ and $s^3 \le 26244\, A$.*

Proof. Split the increments into *runs*: maximal stretches containing no rise. There are at most $\rho + 1$ runs.

(1) By Lemma 9.3, the drops strictly decrease along a run, so each value of d_j occurs at most once per run, hence at most $\rho + 1$ times. Lemma 9.4 and $\sum_j d_j = x_0 - x_{T-1} \le A - 1$ give the claim.

(2) The increments are ≥ 1, and they sum to $q_{T-1} - q_0 \le \lfloor a/A \rfloor$.

(3) Call an increment h *low* if $26244h^3A^4 < a^3$. Five consecutive equal increments h inside the run give six quotients in arithmetic progression. Lemma 9.1 then gives $q^4 \le 324h^3a$. On the other hand, $a < 3Aq$, which follows from $a = qx_0 + x_1$, $x_1 < x_0 < 2A$ and $6A \le a$. Together these give $a^3 \le 26244h^3A^4$, so h is not low. Within a run the increments are nonincreasing, so equal values are consecutive. Hence each low value occurs at most 4 times per run, and at most $4(\rho+1)$ times in all. Lemma 9.4 bounds the number ℓ of low increments by $\ell^2 \le 8(\rho+1)\sum_j k_j \le 8(\rho+1)\lfloor a/A \rfloor$. Let s be the number of high increments. If $s = 0$, its bound is immediate. Otherwise let h_* be their smallest value. Then $sh_* \le \lfloor a/A \rfloor$ and $a^3 \le 26244h_*^3A^4$, so $s^3a^3 \le 26244A^4(a/A)^3 \le 26244Aa^3$. □

Proof of Theorem 1.4. The rises inside a window are rises of the orbit, so $\rho + 1 \le R$. Combining the three parts of Proposition 9.5, as in the proof of Theorem 8.4, gives for the window $[2^k, 2^{k+1})$ (when $6A > a$, part (2) gives $T - 1 \le a/A < 6 \le 30A^{1/3}$)

$$T_k \le 1 + \sqrt{R}\,\min\bigl(\sqrt{8a}\,2^{-k/2},\ \sqrt{2}\,2^{k/2}\bigr) + \min\bigl(a\,2^{-k},\ 30\cdot 2^{k/3}\bigr).$$

Lemma 8.3 gives $\sum_k \min(\sqrt{8a}\,2^{-k/2}, \sqrt{2}\,2^{k/2}) \ll (\sqrt{8a}\sqrt{2})^{1/2} = 2a^{1/4}$. It also gives $\sum_k \min(a2^{-k}, 30\cdot 2^{k/3}) \ll a^{1/4}$. Since $\log_2 a + 1 \ll a^{1/4}$, summing over k gives the theorem. □

Remark 9.6. By Theorem 1.4, an orbit with $P(a,b) \asymp a^{3/10}$ must have $R \gg a^{1/10}$. Proposition 9.5(3) says more: inside the critical window $A \asymp a^{3/5}$, a run of $T \asymp a^{3/10}$ remainders must contain $\gg a^{1/5}$ rises.

10. The Hankel hierarchy

We now index the orbit from zero: $b_0 = b$, $N = P(a,b)$ and $b_0 > \cdots > b_{N-1} > 0 = b_N$. Write $q_i = \lfloor a/b_i \rfloor$ and $k_i = q_{i+1} - q_i$. For each remaining tail we construct a probability measure on its quotient differences. Reweighting by the reciprocal quotient will make its total mass equal to b_i/a.

Put $p_i(j) = q_{i+j} - q_i$ for $0 \le j < N - i$; the symbols p_i, w_i are local to this construction. Define the weights backwards from $w_{N-1}(0) = 1$ by

$$w_i(0) = 1 - \sum_{j=0}^{N-i-2} \frac{w_{i+1}(j)}{p_i(j+1)}, \qquad w_i(j+1) = \frac{w_{i+1}(j)}{p_i(j+1)} \quad (0 \le j < N-i-1).$$

Let μ_i be the measure with mass $w_i(j)$ at $p_i(j)$, and let

$$d\nu_i(x) = \frac{d\mu_i(x)}{q_i + x}, \qquad h_i(s) = \int x^s \, d\nu_i = \sum_{j=0}^{N-i-1} \frac{w_i(j) p_i(j)^s}{q_{i+j}}.$$

The recursion is chosen to reproduce the division equation $q_i h_i(0) + h_{i+1}(0) = 1$ for $i+1 < N$.

Lemma 10.1. (1) *The weights are positive and sum to* 1.
(2) $h_i(0) = b_i/a$, *that is,* $b_i/a = \mathbb{E}\big[1/(q_i + X_i)\big]$ *for* $X_i \sim \mu_i$; *and* $h_{N-1}(s) = 0$ *for* $s \ge 1$.
(3) $h_i(s+1) = \sum_{t \le s} \binom{s}{t} k_i^{s-t} h_{i+1}(t)$ *for* $i + 1 < N$.
(4) $h_i(s) = (-q_i)^s b_i/a + z$ *with* $z \in \mathbb{Z}$.

Proof. (1) Since $p_i(j+1) \ge 1$, the sum defining $w_i(0)$ is at most 1. It is strictly less: if $N - i - 1 \ge 2$, the term $j = 1$ has $p_i(2) \ge 2$; if $i = N - 2$, the last gap satisfies $q_{N-1} \ge q_{N-2} + 2$, because $b_{N-1} \mid a$ gives $q_{N-1} - q_{N-2} - 1 = q_{N-2}(b_{N-2} - b_{N-1})/b_{N-1} \ge 1$. (2) Since $p_i(j+1) = q_{i+j+1} - q_i$, the definitions give $q_i h_i(0) + h_{i+1}(0) = 1$, which is the normalised division equation; and $h_{N-1}(0) = 1/q_{N-1} = b_{N-1}/a$; since $\mu_{N-1} = \delta_0$, also $h_{N-1}(s) = 0$ for $s \ge 1$. (3) Use $p_i(j+1) = k_i + p_{i+1}(j)$ and the binomial theorem. (4) For $i = N - 1$ this holds with $z = -(-q_{N-1})^s b_{N-1}/a = -(-1)^s q_{N-1}^{s-1}$ for $s \ge 1$, since $q_{N-1} b_{N-1} = a$. Then induct backwards on i using (3), $b_{i+1}/a = 1 - q_i b_i/a$ and $k_i - q_{i+1} = -q_i$. □

Theorem 10.2. *For* $1 \le r \le N - i$, *the number* $I_r(i) = a \det\big(h_i(j+l)\big)_{0 \le j,l < r}$ *is a positive integer. Moreover* $I_1(i) = b_i$, $I_2(i) = d_i$ *(if* $i+1 < N$*),* $I_3(i) = D_i$ *(if* $i+3 < N$*), and* $I_4(i) = L_i$ *(if* $i+5 < N$*), where* D_i *and* L_i *are the curvature and the six-step integer of the blocks starting at* b_i.

Proof. Integrality. By Lemma 10.1(4), $h_i(s+1) + q_i h_i(s) \in \mathbb{Z}$. Multiply the Hankel matrix on the left by the unit lower triangular matrix that adds q_i times row $j-1$ to row j. This makes rows $1, \dots, r-1$ integral and leaves the determinant unchanged. Expand along row 0, whose entries are $(-q_i)^l b_i/a + (\text{integer})$; then $a \det$ is an integer.

Positivity. The Hankel matrix is the Gram matrix of $1, x, \dots, x^{r-1}$ in $L^2(\nu_i)$. Its quadratic form is $\sum_j w_i(j) q_{i+j}^{-1} \big(\sum_l c_l p_i(j)^l\big)^2$. If this vanishes, the polynomial $\sum_l c_l x^l$ vanishes at the $N - i \ge r$ distinct points $p_i(j)$, so it is zero, by the invertibility of the Vandermonde matrix. So the matrix is positive definite.

The first four. First, $I_1(i) = a h_i(0) = b_i$. For $i+1 < N$, write $\beta_j = b_{i+j}/a$. Then $h_i(0) = \beta_0$, $h_i(1) = \beta_1$ and $h_i(2) = k_i \beta_1 + \beta_2$. Hence

$$I_2(i) = a\big(\beta_0(k_i\beta_1 + \beta_2) - \beta_1^2\big) = a(\beta_0 - \beta_1) = d_i,$$

using $q_i\beta_0 + \beta_1 = q_{i+1}\beta_1 + \beta_2 = 1$. For the larger determinants, Lemma 10.1(2),(3) makes the entries $h_i(s)$, $s \le 2r - 2$, explicit polynomials in $\beta_j = b_{i+j}/a$ and $k_i, \dots, k_{i+2r-4}$; at the end

of the orbit this uses $h_{N-1}(s) = 0$ for $s \geq 1$, that is, $b_N = 0$. Expanding and reducing by $q_{i+j}\beta_j + \beta_{j+1} = 1$ gives D_i and L_i in the stated ranges. The formal proof records polynomial certificates for these two reductions. □

The ranges $i+3 < N$ and $i+5 < N$ cannot be relaxed to all i at which D_i or L_i is defined. For $a = 926$ and $b = 229$ the orbit is $229, 10, 6, 2$, so $N = 4$, and $I_3(1) = 984$ while $D_1 = 988$.

Remark 10.3. In these index ranges, the positivity of D and L, proved earlier by explicit identities, is an instance of the positivity of Gram determinants. The integrality of L (Lemma 7.1(1)) is an instance of the integrality of I_4. A priori $I_r(i)$ involves the moments $h_i(s)$ for $s \leq 2r-2$, and hence $b_i, \dots, b_{i+2r-2}$. In fact $I_3(i)$ depends only on b_i, b_{i+1}, b_{i+2} and q_i, q_{i+1}, q_{i+2}. Similarly $I_4(i)$ depends only on $b_i, \dots, b_{i+3}$ and $q_i, \dots, q_{i+4}$: the later remainders cancel.

11. Breaking the exponent 3/10

In this section we prove Theorem 1.5. We keep the notation of Section 10: $b_0 > b_1 > \cdots > b_{N-1} > 0$ is the positive part of the orbit, $N = P(a,b)$, and $q_i = \lfloor a/b_i \rfloor$.

11.1. **Heuristics.** The exponent 3/10 of Theorem 1.2 comes from a single balanced configuration: $T \asymp a^{3/10}$ remainders in a window $[A, 2A)$ with $A \asymp a^{3/5}$, quotients $q \asymp a^{2/5}$ and typical increments $k \asymp a^{1/10}$. At this scale the four-term Hankel invariant $I_4(i) = L(i)$ has size $\asymp aH^6/q^4 \asymp 1$, where H bounds the increments. The idea is that a positive integer of bounded size carries information. The labels $I_4(i), I_4(i+1), I_4(i+2)$, together with four further integers of the same size, determine a *pattern*. States with a given pattern satisfy two linear equations and one quadratic equation in five consecutive increments. So a pattern accounts for at most $O(H^2)$ increment tuples, and each tuple for at most four states. At the balanced point this gives $T \ll H^2 \asymp a^{1/5}$, a large saving. Moving away from the balanced point costs a power of the pattern size, and optimising gives the exponent 23/77.

11.2. **The pattern matrix.** Fix a state i with $i+6 < N$ and write, for $m \geq 0$,

$$k_m = q_{i+m+1} - q_{i+m}, \qquad d_m = b_{i+m} - b_{i+m+1},$$
$$s_m = k_0 + \cdots + k_{m-1}, \qquad r_m = k_m(s_m + s_{m+1}) + k_{m+1}.$$

All indices here are local to the chosen state, and the definitions apply whenever the indicated orbit entries exist; the empty sum gives $s_0 = 0$. So $s_m = q_{i+m} - q_i$, $r_0 = k_0^2 + k_1$, $r_1 = k_1(2k_0 + k_1) + k_2$, and so on. Consider the 3×4 integer matrix

$$\mathcal{M}_i = \begin{pmatrix} d_0 & d_1 & d_2 & d_3 \\ k_0 & k_1 & k_2 & k_3 \\ r_0 & r_1 & r_2 & r_3 \end{pmatrix},$$

and let $U_m(i) = (-1)^{m+1} \det(\mathcal{M}_i \text{ with column } m \text{ deleted})$ for $0 \leq m \leq 3$.

Lemma 11.1. (1) *$U_3(i) = L(i) = I_4(i)$ and $U_0(i) = -L(i+1) = -I_4(i+1)$, where L is the six-step integer of Lemma 7.1.*
(2) *$\sum_m U_m(i)\, d_m = \sum_m U_m(i)\, k_m = \sum_m U_m(i)\, r_m = 0$.*

Proof. (2) is the Laplace expansion of a 4×4 determinant with a repeated row. For (1), the identity $\det(\text{columns } 0,1,2) = L(i)$ holds as a polynomial identity in $d_0, d_1, d_2, k_0, \dots, k_3$, by direct expansion; and $L(i) = I_4(i)$ by Theorem 10.2. Passing from i to $i+1$ shifts the d- and k-rows by one column, and, since $s_m(i+1) = s_{m+1}(i) - k_0$, it replaces r_m by $r_{m+1} - 2k_0k_{m+1}$. This row operation does not change the determinant, so $\det(\text{columns } 1,2,3) = L(i+1)$. □

Remark 11.2. The vector $C_m = U_m/U_3$ has a meaning: $\pi_i = \sum_{m\le 3} C_m(F_{m+1} - F_m)$, with Newton polynomials $F_m(x) = \prod_{l<m}(x - s_l)$, is the monic orthogonal polynomial of degree four for the measure ν_i of Section 10. The three rows of $\mathcal{M}_i$ encode $\nu_i(\pi_i) = 0$ and $\mu_i(x\pi_i) = \mu_i(x^2\pi_i) = 0$, and $\nu_i(xf) = \mu_i(f) - q_i\nu_i(f)$ turns these into $\nu_i(x^j\pi_i) = 0$ for $j \le 3$. In particular $U_m(i)/I_4(i)$ are tail sums of the Newton coefficients of π_i: writing $\pi_i = \sum_{l\le 4}\gamma_l F_l$, one has $U_m(i)/I_4(i) = \gamma_{m+1} + \cdots + \gamma_4$, and $\gamma_0 = -C_0 = I_4(i+1)/I_4(i)$. This interpretation is not needed for the elementary count in this section; it will be used in Section 15.

The *pattern* of i is the vector of seven integers

$$\theta(i) = \big(L(i),\, L(i+1),\, L(i+2),\, U_1(i),\, U_2(i),\, U_1(i+1),\, U_2(i+1)\big),$$

and its *gaps* are $\kappa(i) = (k_0, \dots, k_4)$. By Lemma 11.1, $\theta(i)$ determines $U(i)$ and $U(i+1)$ completely.

11.3. **Three consequences.**

Lemma 11.3 (The reduced quadratic)**.** *Write $\theta(i) = (N_0, N_1, N_2, u_1, u_2, v_1, v_2)$ and $\kappa(i) = (k_0, \dots, k_4)$. Then*

$$N_1N_0(N_1 + N_0)\,k_3^2 + \beta\,k_3 + \gamma = 0,$$

where β and γ are polynomials in $\theta(i)$, k_1 and k_2 alone.

Proof. Lemma 11.1(2) for the k-row at i and at $i+1$ gives

$$N_1k_0 = u_1k_1 + u_2k_2 + N_0k_3, \qquad N_1k_4 = N_2k_1 - v_1k_2 - v_2k_3.$$

Substitute these into $N_1^2\sum_m U_m(i)r_m = 0$. Since k_0 enters the r_m at most quadratically and k_4 only linearly, the result is a polynomial in (θ, k_1, k_2, k_3). The coefficient of k_3^2 comes from $-N_1\cdot N_1^2k_0^2$, from $N_0\cdot N_1^2\cdot 2k_0k_3$ and from $N_0\cdot N_1^2k_3^2$, and it equals $-N_1N_0^2 + 2N_1N_0^2 + N_1^2N_0 = N_1N_0(N_1+N_0)$. Explicitly $\beta = 2N_1^2N_0(k_1+k_2) + N_1^2u_2 + 2N_1N_0(u_1k_1 + u_2k_2) - N_1N_0v_2$, and γ is given in the Lean file `TwoState.lean`. □

Lemma 11.4 (The multiplicity identity)**.** *Let $F_m(Y) = \prod_{l<m}(Y - s_l)$ and $\mathcal{Q}_i(Y) = \sum_{m=0}^{3} U_m(i)\,F_m(Y)\,(Y - s_m - 1)$. Then*

$$\mathcal{Q}_i(-q_i) = a\,H_4(k_0, \dots, k_4),$$

where H_4 is an explicit polynomial with 43 terms, independent of i. As a polynomial in q, $\mathcal{Q}_i(-q)$ has degree four and leading coefficient $U_3(i) = L(i) \ge 1$.

Proof. The division equations read $b_{i+m+1} = a - (q_i + s_m)b_{i+m}$. Use them to eliminate $b_{i+1}, \dots, b_{i+4}$ from both sides. The difference becomes the zero polynomial in $a, q_i, b_i, k_0, \dots, k_4$; explicit certificates are in the Lean proof. The polynomial H_4 is the Hankel determinant $\det\big(\mu_i(x^{j+l})\big)_{j,l<4}$, and the identity is $I_4(i)\,\pi_i(-q_i) = a\det\big(\mu_i(x^{j+l})\big)$. This follows from the determinant formula for π_i, since $\nu_i(x^{s+1}) + q_i\nu_i(x^s) = \mu_i(x^s)$. The leading coefficient comes from the term $m = 3$. □

Lemma 11.5 (Labels)**.** *Let $i + 6 < N$ and put*

$$G(k_0, \dots, k_4) = (k_0+1)\big(k_1s_2 + s_2 + k_1 + k_2 + 1\big)\big(k_2(k_1+k_2)s_3 + e_3\big),$$

where $e_3 = s_3(k_1+k_2) + s_3k_2 + (k_1+k_2)k_2 + (k_0 + 2k_1 + 3k_2)(k_3+1) + k_3^2 + 2k_3 + k_4 + 1$. Then $L(i)\,q_i^4 \le a\,G(k_0, \dots, k_4)$. If moreover $b_i < 2A$, then $L(i)\,a^3 \le 256A^4\,G(k_0, \dots, k_4)$.

Proof. The Newton polynomials $F_0, \dots, F_3$ are monic of degrees $0, \dots, 3$. So the matrix $\big(\nu_i(F_jF_l)\big)$ is $\mathcal{U}\mathcal{H}\mathcal{U}^{\mathsf{T}}$, where $\mathcal{H}$ is the Hankel matrix of ν_i and $\mathcal{U}$ is unit lower triangular. Hence it has determinant $I_4(i)/a$, and it is positive definite, as in the proof of Theorem 10.2. By Hadamard's inequality,

$$\frac{I_4(i)}{a} \le \prod_{j<4} \nu_i(F_j^2) \le \prod_{j<4} \frac{\mu_i(F_j^2)}{q_i},$$

since $\nu_i = \mu_i/(q_i + x) \le \mu_i/q_i$ on the nonnegative support. By the shift rule $\mu_i(F_j g) = \mu_{i+j}\big(g(x + s_j)\big)$ and the moments $\mu_{i+m}(1) = \mu_{i+m}(x) = 1$, $\mu_{i+m}(x^2) = k_m + 1$, $\mu_{i+m}(x^3) = k_m^2 + 2k_m + k_{m+1} + 1$, the product equals G. For the second claim, $a = q_ib_i + b_{i+1} < 2A(q_i+1) \le 4Aq_i$, so $L(i)a^4 \le 256A^4L(i)q_i^4 \le 256A^4aG$. □

11.4. **The size of a pattern.**

Lemma 11.6. *Let $1 \le k_0, \dots, k_5 \le H$ and $A \ge 1$. Suppose $N_0 = L(i)$ and $N_1 = L(i+1)$ satisfy $N_0a^3 \le 256A^4G(k_0, \dots, k_4)$ and $N_1a^3 \le 256A^4G(k_1, \dots, k_5)$, and that the* scale condition

$$237568\, A^4H^5 \le a^3 \tag{11}$$

holds. Then $L(i)$, $L(i+1)$, $|U_1(i)|$ and $|U_2(i)|$ are all at most $172032\, A^4H^6/a^3$.

Proof. Put $V = k_1k_2(k_1 + k_2)$ and $\Delta_4 = k_0 \cdot k_1s_2 \cdot k_2(k_1 + k_2)s_3$. Expanding G and bounding each of the three error terms by $56H^5$, $170H^5$ and $238H^5$ gives $G(k_0, \dots, k_4) \le \Delta_4 + 464H^5$. Also $\Delta_4 \le 6H^3V$, because $\Delta_4 = k_0s_2s_3 \cdot V$. Similarly $G(k_1, \dots, k_5) \le 6H^3V + 464H^5$, because the corresponding product is $V \cdot k_3(k_2 + k_3)(k_1 + k_2 + k_3)$. By (11) and $N_0, N_1 \ge 1$, $256A^4 \cdot 464H^5 \le a^3/2 \le N_ja^3/2$ for $j = 0, 1$, so

$$N_0a^3,\ N_1a^3 \le 3072\, A^4H^3V.$$

Since $N_0 \ge 1$, this and (11) give $V \ge 77H^2$.

Now solve for $(u_1, u_2) = (U_1(i), U_2(i))$ from the k- and r-relations of Lemma 11.1. These read $u_1k_1 + u_2k_2 = N_1k_0 - N_0k_3$ and $u_1r_1 + u_2r_2 = N_1r_0 - N_0r_3$. The determinant is

$$\delta = k_1r_2 - k_2r_1 = k_1k_2(k_1 + k_2) + k_1k_3 - k_2^2 \ \ge\ V - H^2 \ \ge\ V/2.$$

Using $r_m \le (2m + 2)H^2$, Cramer's rule gives $|u_j|\delta \le 14(N_0 + N_1)H^3$, hence

$$|u_j|\, Va^3 \le 28H^3(N_0 + N_1)a^3 \le 28H^3 \cdot 6144A^4H^3V,$$

that is, $|u_j|a^3 \le 172032A^4H^6$. Finally $N_ja^3 \le 3072A^4H^3V \le 6144A^4H^6$, since $V \le 2H^3$. □

The point of the argument is in the first paragraph. The two adjacent labels share the Vandermonde factor V, so the ratio $L(i+1)/L(i)$ never enters the bound.

11.5. **The counting lemma.**

Proposition 11.7. *Let $\mathcal{G}$ be a set of states i with $i + 8 < N$. Suppose that for each $i \in \mathcal{G}$ all entries of $\theta(i)$ have absolute value at most M, and all entries of $\kappa(i)$ lie in $[1, H]$. Then $|\mathcal{G}| \le 8(2M + 1)^7H^2$.*

Proof. (i) At most four states per (θ, κ). If $\theta(i) = \theta$ and $\kappa(i) = \kappa$, then by Lemma 11.4 q_i is a root of the polynomial $\mathcal{Q}(-q) - aH_4(\kappa)$. This polynomial depends only on (θ, κ), and it has degree four and leading coefficient $\theta_0 \ge 1$. Since q_i is strictly increasing in i, there are at most four such states.

(ii) At most two values of κ per (θ, k_1, k_2). By Lemma 11.3, k_3 is a root of a quadratic whose coefficients depend only on (θ, k_1, k_2). The leading coefficient is $N_1N_0(N_1 + N_0) > 0$. The two linear relations then determine k_0 and k_4, because $N_1 \ne 0$.

(iii) There are at most $(2M+1)^7 H^2$ values of (θ, k_1, k_2). □

11.6. **The window bound.**

Proposition 11.8. *Let $b_s, \dots, b_{s+T-1}$ be a window run in $[A, 2A)$ with $T \geq 36$, and put $H = \lfloor 28a/(AT) \rfloor + 1$ and $M = \lfloor 172032 A^4 H^6 / a^3 \rfloor$. If (11) holds, then*

$$T \leq 16\,(2M+1)^7 H^2.$$

Proof. Call a start i *good* if $s \leq i$, $i+9 \leq s+T$ and $q_{j+1} - q_j \leq H$ for $i \leq j \leq i+6$, that is, $k_0, \dots, k_6 \leq H$ at state i.

Most starts are good. Let $\mathcal{B}$ be the set of $j \in [s, s+T-1)$ with $q_{j+1} - q_j > H$. Then $|\mathcal{B}|(H+1) \leq q_{s+T-1} - q_s \leq a/A$. Since $H \cdot AT > 28a$, this gives $28|\mathcal{B}| \leq T$. Each $j \in \mathcal{B}$ spoils at most seven starts, so at least $T - 9 - 7|\mathcal{B}| \geq T/2$ starts are good, as $T \geq 36$.

Good starts satisfy Proposition 11.7. Let i be good. Then $i+8 < N$, and $b_i, b_{i+1}, b_{i+2} < 2A$. So Lemma 11.5 applies at i, $i+1$ and $i+2$. Lemma 11.6 at i and at $i+1$ bounds all seven entries of $\theta(i)$ by M. Therefore $T/2 \leq 8(2M+1)^7 H^2$. □

11.7. **Proof of Theorem 1.5.**

Proposition 11.9. *Let $T \geq 36$ be the length of a window run in $[A, 2A)$. Then $T^{77} \leq C_{77} \Delta^{22} a^{23}$, where*

$$C_{77} = 122449^{20} + (8 \cdot 237568 \cdot 56^5)^{11} + (64 \cdot 50176)^{11} + 8^{16} \cdot 16 \cdot 3^7 \cdot 172032^7 \cdot 56^{44}.$$

Proof. We use Proposition 7.3, $T^6 A^2 \leq 96 \cdot 10^6 a^3$, and Proposition 5.1, which gives $T^2 \leq 2T(T-1) \leq 8A\Delta$.

Case $AT > 28a$. Then $784 a^2 T^4 \leq T^6 A^2 \leq 96 \cdot 10^6 a^3$, so $T^4 \leq 122449\, a$ and $T^{77} \leq 122449^{20} a^{20}$.

From now on $AT \leq 28a$, so $H \cdot AT \leq 28a + AT \leq 56a$.

Case (11) fails. Then $a^3 (AT)^5 < 237568 A^4 (H\,AT)^5 \leq 237568 \cdot 56^5 A^4 a^5$, so $AT^5 \leq 237568 \cdot 56^5 a^2$. Hence $T^7 = T^5 T^2 \leq 8\Delta\, AT^5 \leq 8 \cdot 237568 \cdot 56^5\, \Delta a^2$. Raise this to the eleventh power.

Case (11) holds and $M = 0$. Then $T \leq 16H^2$, so $T(AT)^2 \leq 16(56a)^2$. Hence $T^7 = T^3 (T^2)^2 \leq 64 \Delta^2\, T^3 A^2 \leq 64 \cdot 50176\, \Delta^2 a^2$. This case is in fact vacuous: the proof of Proposition 11.8 provides a good start i, and there $1 \leq L(i) \leq M$ by Lemma 11.6. We keep it, here and in the analogous proofs below, because the Lean proofs treat it separately.

Case (11) holds and $M \geq 1$. Then $2M+1 \leq 3M$ and $Ma^3 \leq 172032 A^4 H^6$, so

$$T\, a^{21} \leq 16 \cdot 3^7\, (Ma^3)^7 H^2 \leq 16 \cdot 3^7 \cdot 172032^7\, A^{28} H^{44}.$$

Multiplying by $(AT)^{44}$ and using $H \cdot AT \leq 56a$ gives $T^{45} A^{16} \leq 16 \cdot 3^7 \cdot 172032^7 \cdot 56^{44}\, a^{23}$. Finally

$$T^{77} = T^{45} (T^2)^{16} \leq 8^{16} \Delta^{16}\, T^{45} A^{16} \leq 8^{16} \cdot 16 \cdot 3^7 \cdot 172032^7 \cdot 56^{44}\, \Delta^{16} a^{23}.$$

In every case $T^{77} \leq C_{77} \Delta^{22} a^{23}$. □

Proof of Theorem 1.5. This is the argument of Theorem 1.2, with Proposition 11.9 in place of Corollary 5.3. Let $\varepsilon > 0$ and choose an integer $m > 1/\varepsilon$. By Lemma 2.2, $\Delta^{22m} \leq C_{22m} \cdot 32 a^7$, so every dyadic window has $T \leq 35$ or

$$T^{77m} \leq C_{77}^m\, C_{22m}\, 32\; a^{23m+7}.$$

Since $(23m+7)/(77m) = 23/77 + 1/(11m) \leq 23/77 + \varepsilon/2$, each of the $\lfloor \log_2 a \rfloor + 1$ windows has $T \ll_\varepsilon a^{23/77 + \varepsilon/2}$, and the theorem follows. □

Remark 11.10. In terms of the scales, Proposition 11.8 says $T \ll X^7 H^2$, where $X = A^4 H^6/a^3$ bounds the seven pattern integers and $H \asymp a/(AT)$. Thus $X \asymp a^3/(A^2T^6)$ and $T^{45}A^{16} \ll a^{23}$. The drop bound $A \gg T^2/\Delta$ then gives $T^{77} \ll \Delta^{16}a^{23}$. The scale condition (11) says $Q^4 \gg aH^5$, where $Q = a/A$, and its failure gives the stronger bound $T \ll a^{2/7+o(1)}$.

12. The ternary quadratic and the exponent 11/37

Proposition 11.7 uses only one coefficient of the reduced quadratic. Here we use all of it, which saves a factor H in the count per pattern and proves Theorem 1.6. We keep the notation of Section 11, and, as in Lemma 11.3, write a pattern as $\theta = (N_0, N_1, N_2, u_1, u_2, v_1, v_2) = (L(i), L(i+1), L(i+2), U_1(i), U_2(i), U_1(i+1), U_2(i+1))$. In particular A is still the scale of the window $[A, 2A)$.

12.1. **Signs.** Throughout this subsection the pattern and gaps arise from the same state, so the cofactor relations of Lemma 11.1 remain in force. Put $S(a,b,c) = a^2b + (a-1)b^2 + ac$ and $E(a,b,c,d) = a^2c + (2a-1)bc + ac^2 + ad$. Both are positive at positive integers.

Lemma 12.1 (Signs)**.** *If $N_0, N_1 \geq 1$ and $k_0, \dots, k_4 \geq 1$, then $u_1 > 0 > u_2$.*

Proof. Solving the k- and r-relations of Lemma 11.1 for (u_1, u_2) by Cramer's rule gives the exact identities

$$u_1 S(k_1,k_2,k_3) = N_1 E(k_0,k_1,k_2,k_3) + N_0 S(k_2,k_3,k_4),$$
$$-u_2 S(k_1,k_2,k_3) = N_1 S(k_0,k_1,k_2) + N_0 E(k_1,k_2,k_3,k_4),$$

since the Cramer determinant is $k_1 r_2 - k_2 r_1 = S(k_1,k_2,k_3)$. □

12.2. **The ternary relation.** For a pattern θ let

$$F_\theta(x,y,z) = (u_1x + u_2y + N_0z)^2 + N_1u_1x^2 + N_1u_2y^2 + N_0N_1z^2 + 2N_1u_2xy + 2N_0N_1xz$$
$$+ 2N_0N_1yz + (N_0N_2 - N_1^2)x + (N_1u_1 - N_0v_1)y + (N_1u_2 - N_0v_2)z.$$

Lemma 12.2. *At every state i with $i+7 < N$, $F_{\theta(i)}(k_1,k_2,k_3) = 0$. Write $F_\theta = \alpha x^2 + \beta xy + \gamma y^2 + d(z)x + e(z)y + f(z)$. Then $\alpha = u_1(u_1+N_1)$, $\beta^2 - 4\alpha\gamma = -4N_1u_2(N_1+u_1)(u_1-u_2)$, and*

$$(\beta d_1 - 2\alpha e_1)^2 + (\beta^2 - 4\alpha\gamma)(4\alpha f_2 - d_1^2) = 16\, N_0N_1^2u_1(N_0-u_2)(N_1+u_1)^2(u_1-u_2),$$

where d_1, e_1, f_2 are the coefficients of z, z, z^2 in d, e, f, respectively. In particular, under the hypotheses of Lemma 12.1, $\alpha > 0$, $\beta^2 - 4\alpha\gamma > 0$ and the last quantity is nonzero.

Proof. The identity $N_1F_\theta(k_1,k_2,k_3) = N_1^2R + (N_0N_1k_3 - N_1^2k_0 + N_1u_1k_1 + N_1u_2k_2)K_0 - N_0N_1K_1$, where K_0, K_1 and R are the k-relations at i, $i+1$ and the r-relation at i, gives the first claim. The rest is expansion. So $F_\theta \equiv N_1R$ modulo the two linear relations. □

12.3. **Norm equations.**

Lemma 12.3 (Norm equations)**.** *Let $D \geq 1$, $m \neq 0$, $B \geq 1$ be integers and $Z = B(\lfloor\sqrt{D}\rfloor + 2)$. The number of $(x,y) \in [-B,B]^2$ with $x^2 - Dy^2 = m$ is at most $2\tau(|m|)^3(\log_3 Z^2 + 1)$.*

Proof. If $D = h^2$ then $(x-hy)(x+hy) = m$, and $x - hy$ is one of the $2\tau(|m|)$ signed divisors of m; it determines (x,y). Otherwise write $D = df^2$ with $d > 1$ squarefree, so d is not a square and $\sqrt{d}$ is irrational. For a solution put $g = \gcd(x, fy)$, $u = x/g$, $w = fy/g$ and $n = u^2 - dw^2$, so that $m = g^2n$. Since $\gcd(u,w) = 1$ and $n \equiv u^2 \pmod w$, w is invertible modulo n. Let $t \in [0, |n|)$ be the residue of uw^{-1}; then $t^2 \equiv d$ and $u \equiv tw$ modulo n.

The residues. The number of square roots of d modulo $c \geq 1$ is at most $\tau(c)^2$. Modulo p^e it is at most four. If $p \mid d$, either $e = 1$ and the only root is 0, or $e \geq 2$ and a root would force $p^2 \mid d$. If $p \nmid d$ is odd, two roots r, s satisfy $p^e \mid (s-r)(s+r)$, and p cannot divide both factors, so $s \equiv \pm r$. If $p = 2 \nmid d$, then s, r are odd, one of $(s \mp r)/2$ is odd, and $s \equiv \pm r \pmod{2^{e-1}}$. The count is submultiplicative over coprime factors, by the Chinese remainder theorem, and $4 \leq \tau(p^e)^2$.

A class. Fix g (with $g^2 \mid m$), t and the sign of $\xi = u + w\sqrt{d}$. Two solutions $(u, w), (u', w')$ in this class satisfy $n \mid uu' - dww'$ and $n \mid wu' - uw'$. So $\xi = \xi'\varepsilon$ with $\varepsilon = r + s\sqrt{d}$, where $r = (uu' - dww')/n$, $s = (wu' - uw')/n$ and $r^2 - ds^2 = 1$. If $|\xi| > |\xi'|$ then $\varepsilon > 1$, and then $\varepsilon \geq 3$: indeed $\varepsilon^{-1} = r - s\sqrt{d} \in (0, 1)$, so $2r = \varepsilon + \varepsilon^{-1} > 2$, hence $r \geq 2$, and $\varepsilon < 3$ would give $\varepsilon + \varepsilon^{-1} < 4$. Now $|u| + |w|\sqrt{d} \leq |x| + |y|\sqrt{D} \leq Z$, so $|\xi|, |\bar{\xi}| \leq Z$, and $|\xi\bar{\xi}| = |n| \geq 1$ gives $Z|\xi| \geq 1$. Hence the map $\xi \mapsto \lfloor \log_3(Z|\xi|) \rfloor$ is injective on the class, with values in $[0, \log_3 Z^2]$.

There are at most $\tau(|m|)$ values of g, $\tau(|m|)^2$ of t and two signs. □

The argument is the classical one; compare Vaughan and Wooley [14, Lemma 3.5].

12.4. **Slices.**

Lemma 12.4 (Slices)**.** *Let $F(x, y, z) = \alpha x^2 + \beta xy + \gamma y^2 + (d_1 z + d_0)x + (e_1 z + e_0)y + f_2 z^2 + f_1 z + f_0$ have integer coefficients bounded by K, with $\alpha > 0$, $D = \beta^2 - 4\alpha\gamma > 0$ and $\ell = (\beta d_1 - 2\alpha e_1)^2 + D(4\alpha f_2 - d_1^2) \neq 0$. Let $H \geq 1$, and suppose that $2\tau(|m|)^3(\log_3\big((11K^2H(3K+2))^2\big) + 1) \leq W$ for all integers $0 < |m| \leq 116K^4H^2$. Then F has at most $(2H+1)(W+4)$ zeros in $[-H, H]^3$.*

Proof. Put $t = 2\alpha x + \beta y + d(z)$, $\Lambda(z) = \beta d(z) - 2\alpha e(z)$, $s = Dy + \Lambda(z)$ and $\mathcal{N}(z) = \Lambda(z)^2 + D(4\alpha f(z) - d(z)^2)$. Completing squares gives $s^2 - Dt^2 = \mathcal{N}(z) - 4\alpha DF$. In the box, $|s|, |t| \leq 11K^2H$, $\sqrt{D} \leq 3K$ and $|\mathcal{N}(z)| \leq 116K^4H^2$. For fixed z the map $(x, y) \mapsto (s, t)$ is injective. So if $\mathcal{N}(z) \neq 0$ the slice has at most W zeros, by Lemma 12.3. The z^2-coefficient of $\mathcal{N}$ is $\ell \neq 0$, so at most two slices have $\mathcal{N}(z) = 0$. In such a slice $s^2 = Dt^2$, which for fixed y is a quadratic in x with leading coefficient $4\alpha^2 D \neq 0$, so it has at most $2(2H+1)$ zeros. □

12.5. **The count.**

Proposition 12.5. *Under the hypotheses of Proposition 11.7, let $K = 4M^2$, and suppose W satisfies the hypothesis of Lemma 12.4. Then $|\mathcal{G}| \leq 4(2M+1)^7(2H+1)(W+4)$.*

Proof. As before, at most four states share (θ, κ), and κ is determined by (θ, k_1, k_2, k_3). For a fixed θ, the triples (k_1, k_2, k_3) are zeros of F_θ in $[1, H]^3$ (Lemma 12.2). The coefficients of F_θ are at most $4M^2$, and Lemmas 12.1 and 12.2 give the nondegeneracy hypotheses of Lemma 12.4. □

Let $P_1(a) = 10^{300}a^{100}$, let $\tau^*(X) = \max_{n \leq X} \tau(n)$ with the convention $\tau(0) = 0$, and let $\Xi(a) = 2\tau^*(P_1(a))^3(\log_3 P_1(a) + 1) + 4$. In a window $H \leq 29a$ and $M \leq 10^{16}a^7$, also for the larger M of Proposition 13.4 below, so $116K^4H^2 \leq P_1(a)$ and $(11K^2H(3K+2))^2 \leq P_1(a)$. Thus $W = \Xi(a) - 4$ is admissible, and the argument of Proposition 11.8 gives:

Proposition 12.6. *In the setting of Proposition 11.8, if* (11) *holds then $T \leq 24(2M+1)^7 H\,\Xi(a)$.*

Proposition 12.7. *For every window run with $T \geq 36$, $T^{74} \leq C_{74}\Delta^{19}\Xi(a)^{19}a^{22}$, where $C_{74} = 122449^{19} + (8 \cdot 237568 \cdot 56^5)^{11} + 10752^{19} + 8^{15} \cdot 24 \cdot 3^7 \cdot 172032^7 \cdot 56^{43}$.*

Proof. The four cases of Proposition 11.9 carry over. If $AT > 28a$, then $T^4 \leq 122449a$. If the scale condition fails, then $T^7 \leq 8 \cdot 237568 \cdot 56^5 \Delta a^2$. If $M = 0$, then $T^2 A \leq 1344\,\Xi a$ and

$T^4 \le 10752\,\Delta\Xi a$. If $M \ge 1$, then $Ta^{21} \le 24 \cdot 3^7 \cdot 172032^7 A^{28} H^{43}\Xi$, so $T^{44}A^{15} \le 24 \cdot 3^7 \cdot 172032^7 \cdot 56^{43}\,\Xi a^{22}$, and $T^{74} = T^{44}(T^2)^{15} \le (8A\Delta)^{15}T^{44}$. □

Proof of Theorem 1.6. By the divisor bound $\tau(n)^k \le C_k n$, both Δ and $\tau^*(P_1(a))$ are $O_\eta(a^\eta)$, and $\log_3 P_1(a) \ll_\eta a^\eta$. So $\Xi(a) \ll_\eta a^\eta$, and Proposition 12.7 gives $T \ll_\eta a^{(22+38\eta)/74}$ for every dyadic window. Since $22/74 = 11/37$, summing over the $O(\log a)$ windows proves the theorem. □

Remark 12.8. In terms of the scales, the count per pattern is now $O(Ha^\eta)$ instead of $O(H^2)$. So $T \ll a^\eta X^7 H$, which gives $T^{44} \ll a^{7+\eta}Q^{15}$, and with $Q \ll a^{1+\eta}/T^2$ the bound $T^{74} \ll a^{22+O(\eta)}$. At the critical point $T \approx a^{11/37}$, $Q \approx a^{15/37}$, $H \approx a^{4/37}$ and $X \approx a^{1/37}$.

13. Ten minors and the exponent 8/27

The seven pattern integers of Section 12 were counted as if they were independent. They are not: they are coordinates of a single vector of 3×3 minors. Counting such vectors directly saves a factor X^2 and proves Theorem 1.7.

13.1. **The full minor vector.** Let $\mathcal{M}_i$ now be the 3×5 matrix with rows $(d_0, \dots, d_4)$, $(k_0, \dots, k_4)$ and $(r_0, \dots, r_4)$, and let $p_{c_1c_2c_3}$ be its minor on columns $c_1 < c_2 < c_3$. Expanding the cofactors of Section 11 gives

$$\theta(i) = (p_{012},\, p_{123},\, p_{234},\, p_{023},\, -p_{013},\, p_{134},\, -p_{124}).$$

Lemma 13.1 (Expansion)**.** *Let $q = q_i$ and $s_j = q_{i+j} - q$. If $i + j + 3 < N$ then*

$$\frac{b_{i+j}}{a} = \frac{1}{q} - \frac{1+s_j}{q^2} + \frac{s_j^2 + 2s_j + k_j + 1}{q^3} - e_j, \qquad 0 \le e_j \le \frac{\mu_{i+j}\big((x+s_j)^3\big)}{q^4}.$$

Proof. Apply μ_{i+j} to $\frac{1}{q+y} = \frac{1}{q} - \frac{y}{q^2} + \frac{y^2}{q^3} - \frac{y^3}{q^3(q+y)}$ with $y = s_j + x \ge 0$, using $b_{i+j}/a = \nu_{i+j}(1) = \mu_{i+j}\big(1/(q+s_j+x)\big)$, $\mu(1) = \mu(x) = 1$ and $\mu(x^2) = k+1$. □

Proposition 13.2 (All minors)**.** *Let $i + 8 < N$, $b_i < 2A$ and $k_0, \dots, k_6 \le H$ at state i. Then every minor of $\mathcal{M}_i$ satisfies $|p|\,a^3 \le 3609600\,A^4H^6$.*

Proof. Subtracting the expansions for j and $j+1$ gives $d_j/a = k_j/q^2 - (r_j + k_j)/q^3 + e_{j+1} - e_j$. So replacing the first row by $d'_j = d_j - a(q^{-2} - q^{-3})k_j + aq^{-3}r_j = a(e_{j+1} - e_j)$ changes no minor. Now $|d'_j| \le 235aH^3/q^4$, $|k_j| \le H$ and $|r_j| \le 10H^2$, so $|p| \le 14100\,aH^6/q^4$. Finally $a \le 4Aq$. □

This replaces Lemma 11.6, and the scale condition (11) is no longer needed.

13.2. **Counting minor vectors.**

Proposition 13.3 (Minor vectors)**.** *The number of vectors of minors of integer 3×5 matrices with all coordinates in $[-Y, Y]$ is at most $1 + 20\sum_{m \le Y} 729\,m^4\tau(m) \ll Y^{5+o(1)}$.*

Proof. For a nonzero vector p choose a pivot π with $|p_\pi| = |m|$ maximal, and let c_4, c_5 be the two columns outside π. Let u_a (v_a) be the minor with the a-th pivot column replaced by c_4 (c_5), with the sign of the column permutation. By Cramer's rule and the Plücker relations, the remaining three minors satisfy $m\,p_{w_a} = \pm(u \times v)_a$ (thirty polynomial identities, checked symbolically). So p is determined by $(\pi, \operatorname{sgn} m, |m|, u, v)$, where $|u_a|, |v_a| \le |m|$ and $|m|$ divides every coordinate of $u \times v$.

Fix $m \ge 1$ and u, and let $g = \gcd(m, u_1, u_2, u_3)$, $n = m/g$, $u = gu_0$. By Bézout there are $c \in \mathbb{Z}^3$ and $x \in \mathbb{Z}$ with $c \cdot u_0 + xn = 1$. If $m \mid u \times v$, then $n \mid u_0 \times v$, and the identity

$v-(c\cdot v)u_0 = xn\,v + c\times(v\times u_0)$ shows $v\equiv(c\cdot v)u_0 \pmod n$. Hence v lies in one of n classes modulo n, each meeting $[-m,m]^3$ in at most $(3g)^3$ points: at most $27mg^2$ choices. Summing over u, grouped by $g=d\mid m$, gives $\sum_u g^2 \le \sum_{d\mid m} d^2(3m/d)^3 \le 27m^3\tau(m)$, so at most $729m^4\tau(m)$ pairs (u,v). There are ten pivots and two signs. □

13.3. **The bound.**

Proposition 13.4. *For a window run in $[A,2A)$ with $T\ge 36$, let $H=\lfloor 28a/(AT)\rfloor+1$ and $M=\lfloor 3609600A^4H^6/a^3\rfloor$. Then $T\le 24\cdot 14581\,(M+1)^5H\,\Xi(a)^2$, and*

$$T^{54}\le C_{54}\Delta^{14}\Xi(a)^{28}a^{16}$$

with $C_{54}=122449^{14}+(448\cdot 24\cdot 14581)^{14}+8^{11}\cdot 32\cdot 24\cdot 14581\cdot 3609600^5\cdot 56^{31}$.

Proof. As in Propositions 11.8 and 12.5, at least $T/2$ starts are good, at most four good starts share (θ,κ), and there are at most $(2H+1)(W+4)$ gap triples per pattern. Since θ is a function of the minor vector, there are at most $1+14580M^5\tau^*(M)\le 14581(M+1)^5\Xi(a)$ patterns, by Propositions 13.2 and 13.3. For the second bound, if $AT>28a$ then $T^4\le 122449a$. Otherwise $HAT\le 56a$. If $M=0$ then $T^4\le 448\cdot 24\cdot 14581\,\Delta\Xi^2a$. Otherwise $Ta^{15}\le 32\cdot 24\cdot 14581\cdot 3609600^5A^{20}H^{31}\Xi^2$, hence $T^{32}A^{11}\ll\Xi^2a^{16}$ and $T^{54}\le(8A\Delta)^{11}T^{32}$. □

Proof of Theorem 1.7. Since $\Delta,\Xi(a)\ll_\eta a^\eta$, every window has $T\ll a^{(16+42\eta)/54}$, and $16/54=8/27$. Choose η sufficiently small in terms of ε, include the bounded-length windows, and sum over the $O(\log a)$ dyadic windows. □

Remark 13.5. In terms of the scales, $T\ll a^\eta X^5H$, so $T^{32}\ll a^{5+\eta}Q^{11}$ and $T^{54}\ll a^{16+O(\eta)}$. At the critical point $T\approx a^{8/27}$, $Q\approx a^{11/27}$, $H\approx a^{3/27}$ and $X\approx a^{1/27}$.

14. Lattices of minor vectors and the exponent 45/152

Proposition 13.4 allows $(2H+1)(W+4)$ gap vectors for every minor vector. But the gap vector of a state is not an arbitrary point of $[1,H]^5$: it lies in a rank-three lattice determined by its minor vector. If that lattice has no short vectors, the slices of the ternary quadratic are few. If it has a short vector, the minor vector satisfies an extra congruence, and few minor vectors do. Balancing the two proves Theorem 1.8. The argument uses only integer arithmetic. Instead of the geometry of numbers it uses a reduced basis obtained by minimising determinants and lengths directly.

14.1. **The lattice of a minor vector.** Let $p\ne 0$ be the minor vector of a 3×5 integer matrix $\mathcal{M}$, with pivot π, $m=p_\pi$, pivot columns $c_1<c_2<c_3$, other columns $c_4<c_5$, and u, v as in Proposition 13.3. Let $\Lambda_p\subseteq\mathbb{Z}^3$ be the subgroup of x with $m\mid u\cdot x$ and $m\mid v\cdot x$. It contains $m\mathbb{Z}^3$. Let $\lambda(p)$ be the least sup-norm of a nonzero vector of Λ_p. Subscripts on L_c and p_{abc} retain the original column labels $0,\dots,4$; c_1,c_2,c_3 enumerate the chosen pivot. We use $\gcd(n,x)$ for $\gcd(n,x_1,x_2,x_3)$ when $x\in\mathbb{Z}^3$.

Lemma 14.1 (Lifting)**.** *Let L_c be me_a if $c=c_a$ is a pivot column, u if $c=c_4$ and v if $c=c_5$. For every row x of $\mathcal{M}$ and every column c, $m\,x_c=L_c\cdot(x_{c_1},x_{c_2},x_{c_3})$. Moreover $\det(L_1,L_2,L_3)=m^2p_{123}$.*

Proof. The rows L_c are the rows of $\mathcal{M}^{\mathsf T}\operatorname{adj}(\mathcal{M}_\pi^{\mathsf T})$, where $\mathcal{M}_\pi$ is the pivot 3×3 block. This is Cramer's rule. Both claims are polynomial identities in the fifteen entries, checked symbolically for each of the ten pivots. □

So the pivot gaps $k'=(k_{c_1},k_{c_2},k_{c_3})$ of a state lie in Λ_p. Its five gaps are $k_c=L_c\cdot k'/m$, and since $p_{123}=L(i+1)\ge 1$, the gaps (k_1,k_2,k_3) are an invertible linear image of k'.

Lemma 14.2 (Index)**.** *If t is a nonnegative integer and $(t+1)^3 > |m|$, then $\lambda(p) \le t$.*

Proof. Write $g = \gcd(|m|, u)$, $n = |m|/g$ and $u = gu_0$. The proof of Proposition 13.3 gives $v = \mu u_0 + nw$ with $\mu = c\cdot v \in \mathbb{Z}$ and $w \in \mathbb{Z}^3$. Put $\psi(x) = \big(u_0\cdot x \bmod n,\ (\mu\lfloor u_0\cdot x/n\rfloor + w\cdot x) \bmod g\big)$, which takes at most $ng = |m|$ values. Two points $x \ne y$ of $[0,t]^3$ with $\psi(x) = \psi(y)$ exist. Then $u_0\cdot(x-y) = n(q-q')$ with q, q' the two floors, so $u\cdot(x-y) = \pm m(q-q')$, and $v\cdot(x-y) = n\big(\mu(q-q') + w\cdot(x-y)\big)$ with the bracket divisible by g. So $x - y \in \Lambda_p$. □

Since $\psi(x) = \psi(y)$ forces $x - y \in \Lambda_p$, the same argument shows that Λ_p has index at most $|m|$ in $\mathbb{Z}^3$.

14.2. **Reduced bases.** Here $|v|$ denotes the Euclidean norm, while $|v|_\infty$ denotes the sup norm used to define $\lambda(p)$. In particular, $|v| \ge |v|_\infty$.

Lemma 14.3 (Reduction)**.** *Let $\Lambda \subseteq \mathbb{Z}^3$ be a subgroup containing $m\mathbb{Z}^3$, $m \ge 1$. There is a basis b_1, b_2, b_3 of Λ with $\delta = \det(b_1, b_2, b_3)$, $0 < |\delta| \le m^3$ and*

$$27\,|b_1|^2|b_2|^2|b_3|^2 \le 64\,\delta^2.$$

Hence every $w \in \Lambda$ is $\sum y_i b_i$ with $y \in \mathbb{Z}^3$ and $27|b_i|^2 y_i^2 \le 64|w|^2$.

Proof. Let I be the least nonzero $|\det|$ of a triple in Λ; the triple (me_1, me_2, me_3) gives $I \le m^3$. A triple with $|\det| = I$ is a basis. Otherwise some $w \in \Lambda$ has a Cramer coordinate that is not an integer, say the first, $\det(w, b_2, b_3)/\delta$, and $w - \lfloor\cdot\rfloor b_1$ gives a triple with smaller nonzero determinant. Among such bases minimise $d_1 = |b_1|^2$, and then the Gram determinant $d_2 = d_1|b_2|^2 - \langle b_1, b_2\rangle^2$. Put $d_3 = \delta^2$.

Choose n with $|2(\langle b_1, b_2\rangle + nd_1)| \le d_1$. Then $(b_2 + nb_1, b_1, b_3)$ is a basis, so $d_1 \le |b_2 + nb_1|^2$, and $d_1|b_2 + nb_1|^2 = d_2 + (\langle b_1, b_2\rangle + nd_1)^2$. Hence $3d_1^2 \le 4d_2$.

Let $B(y,z) = d_1\langle y, z\rangle - \langle b_1, y\rangle\langle b_1, z\rangle$, so that $B(b_2, b_2) = d_2$, and Sylvester's identity gives $B(b_2,b_2)B(b_3,b_3) - B(b_2,b_3)^2 = d_1 d_3$. Choose n_2 with $|2(B(b_2,b_3) + n_2 d_2)| \le d_2$ and put $c = b_3 + n_2 b_2$. Then (b_1, c, b_2) is a basis with first vector b_1, so $d_2 \le d_1|c|^2 - \langle b_1, c\rangle^2 = B(c,c)$, while $d_2 B(c,c) = (B(b_2,b_3) + n_2 d_2)^2 + d_1 d_3$. Hence $3d_2^2 \le 4d_1 d_3$.

Finally choose n_1 with $|2(\langle b_1, c\rangle + n_1 d_1)| \le d_1$, and use the basis b_1, $b_2' = b_2 + nb_1$, $b_3' = c + n_1 b_1$. The identities above give $3d_1|b_2'|^2 \le 4d_2$ and $4d_1 d_2|b_3'|^2 \le d_2^2 + 4d_1 d_3 + d_1^2 d_2$. With the two inequalities this gives $9d_2|b_3'|^2 \le 16 d_3$, and the product of the three bounds is the claim. For the coordinates, $y_1\delta = \det(w, b_2, b_3)$ and Hadamard's inequality $\det(w, b_2, b_3)^2 \le |w|^2|b_2|^2|b_3|^2$. □

14.3. **Few minor vectors have a short vector.**

Lemma 14.4. *For $n \ge 1$ and $x \in \mathbb{Z}^3$, at most $27n^2\gcd(n,x)$ vectors $w \in [-n,n]^3$ have $n \mid w\cdot x$.*

Proof. Reducing modulo n loses a factor at most 27. Let K be the set of solutions in $[0,n)^3$, $e = \gcd(n,x)$ and $c\cdot x \equiv e \pmod n$ by Bézout. The map $(w,j) \mapsto (w + jc) \bmod n$ is injective on $K \times [0, n/e)$: equal images give $n \mid (j - j')e$, so $j = j'$, and then $w = w'$. Hence $|K|\, n/e \le n^3$. □

Proposition 14.5 (Short vectors)**.** *Let $\mathcal{S}$ be a set of nonzero minor vectors with coordinates in $[-Y, Y]$, and let $\tau(m) \le \tau_0$ for $m \le Y$. Then for every $R \ge 1$,*

$$\#\{p \in \mathcal{S} : \lambda(p) \le R\} \le 393660\, R^3 Y^4 \tau_0^2.$$

Proof. If $\lambda(p) \le R$, some nonzero $x \in [-R, R]^3$ has $m \mid u\cdot x$. For fixed x, the encoding of Proposition 13.3 gives at most $20\sum_m N(m,x)$ such p, where $N(m,x)$ counts pairs (u,v) with $m \mid u \times v$ and $m \mid u\cdot x$. For each u there are at most $27m\gcd(m,u)^2$ vectors v. Grouping the u by $g = \gcd(m,u)$ and writing $u = gu'$ with $(m/g) \mid u'\cdot x$, Lemma 14.4 gives $\sum_u g^2 \le$

$27m^2\tau(m)\gcd(m,x)$, so $N(m,x) \le 729m^3\tau(m)\gcd(m,x)$. Finally $\sum_{0<|x|_\infty\le R}\gcd(m,x) \le 27R^3\tau(m)$. A divisor $h > R$ of m divides no nonzero x in the box, and for $h \le R$ the multiples of h number at most $(2R/h+1)^3$, with $h(2R/h+1) \le 3R$. □

14.4. **One minor vector.**

Lemma 14.6 (Definite norm equations)**.** *For $D \ge 1$ and $m \ne 0$, the equation $x^2 + Dy^2 = m$ has at most $4\tau(|m|)^3$ integer solutions.*

Proof. As in Lemma 12.3, with $D = df^2$, d squarefree, each solution reduces to a primitive $u^2 + dw^2 = n$, $m = g^2n$, with a class $t \equiv uw^{-1}$, a square root of $-d$ modulo n. The count of square roots in the proof of Lemma 12.3 applies to $-d$ unchanged. Two solutions in a class give integers r, s with $r^2 + ds^2 = 1$, of which there are at most four. □

A ternary quadratic F with quadratic part of Gram matrix S has $\det(2S)$ multiplied by $\det(C)^2$ under a substitution $x = Cy$. In the notation of Lemma 12.4 one has the identity $\ell = -2\alpha\det(2S)$. For F_θ, Lemma 12.2 therefore gives $\det(2S) = -8N_0N_1^2(N_0 - u_2)(N_1 + u_1)(u_1 - u_2) \ne 0$ under the hypotheses of Lemma 12.1.

If $\det(2S) \ne 0$, a product of two substitutions, each with entries in $\{0, \pm1\}$, determinant ±1 and an inverse with absolute row sums at most 2, makes both the y_1^2 coefficient α and the binary discriminant in (y_1, y_2) nonzero, and then $\ell \ne 0$ as well. First choose one of the planes $\langle e_1, e_2\rangle$, $\langle e_1, e_3\rangle$, $\langle e_2, e_3\rangle$, $\langle e_1 + e_2, e_3\rangle$: if all four binary discriminants vanished then $\det(2S)$ would vanish, by an explicit combination with polynomial coefficients. Writing f, g for the displayed spanning vectors of the chosen plane, choose f, g or $f + g$ as its first basis vector. With this, Lemma 12.4 holds for $\alpha \ne 0$ and $D \ne 0$ of either sign, under the same hypothesis on W. When $D < 0$ each slice uses Lemma 14.6, whose bound $4\tau(|m|)^3$ is at most $2\tau(|m|)^3(\log_3\big((11K^2H(3K+2))^2\big) + 1)$ because $11K^2H(3K+2) \ge 3$.

Proposition 14.7 (One minor vector)**.** *Let p be a minor vector with all coordinates in $[-X, X]$, $X \ge 1$, and suppose W satisfies the hypothesis of Lemma 12.4 for $K = 248832X^8$ and $12H+1$ in place of H. The good starts with minor vector p have at most*

$$\big(8\lfloor 3H/\lambda(p)\rfloor + 3\big)(W+4)$$

gap vectors κ.

Proof. At a good start the gaps lie in $[1, H]$, and since $N_0 = L(i)$ and $N_1 = L(i+1)$ are positive integers, Lemma 12.1 gives $u_1 > 0 > u_2$. Let b_1, b_2, b_3 be the basis of Lemma 14.3 for Λ_p, with $|m|$ in place of m, and $\hat b_i$ their lifts $c \mapsto L_c \cdot b_i/m$ to $\mathbb{Z}^5$. Write $k' = \sum y_ib_i$. By Lemma 14.1, $\kappa = \sum y_i\hat b_i$, so $(k_1, k_2, k_3) = \hat B y$ for the 3×3 matrix $\hat B$ of the lifts in columns $1, 2, 3$, and $m\det\hat B = p_{123}\delta \ne 0$. So y is a zero of $\tilde F = F_\theta \circ \hat B$, and $\det(2S_{\tilde F}) = \det(2S_{F_\theta})(\det\hat B)^2 \ne 0$. Each $|b_i|^2 \ge \lambda(p)^2$ and $|k'|^2 \le 3H^2$, so $\lambda(p)|y_i| \le 3H$, and after the plane substitution the coordinates lie in $[-Y', Y']$ with $Y' = 4\lfloor 3H/\lambda(p)\rfloor + 1 \le 12H + 1$. Every nonzero integer basis vector has length at least one, so Lemma 14.3 gives $|b_i| \le 2X^3$. Maximality of the pivot bounds each entry of L_c by $|m|$. Thus every lift is at most $6X^3$, and the coefficients of the final quadratic are at most $12^3 \cdot 4X^2 \cdot (6X^3)^2 = 248832X^8$. The slice count applies with $(2Y'+1)(W+4)$, and y determines κ. □

14.5. **The bound.**

Lemma 14.8 (Partial summation)**.** *Let t and R_p for $p \in \mathcal{S}$ be integers with $1 \le R_p \le t$, and let $\#\{p : R_p \le r\} \le Cr^3$ for $r \ge 1$. Then $t\sum_p\lfloor n/R_p\rfloor \le n|\mathcal{S}| + nCt^3$ for every $n \ge 0$.*

Proof. Write $1/r = 1/t + \sum_{r\le j<t} 1/(j(j+1))$ and exchange the sums: $\sum_p 1/R_p \le |\mathcal{S}|/t + \sum_{j<t} Cj^3/(j(j+1)) \le |\mathcal{S}|/t + Ct^2$. □

Let $P_2(a) = 10^{1000}a^{400}$ and $\Xi_2(a) = 2\tau^*(P_2(a))^3(\log_3 P_2(a)+1)+4$. As in Section 12, $W = \Xi_2(a) - 4$ is admissible in every window, and $\Xi_2(a) \ll_\eta a^\eta$.

Proposition 14.9. *For a window run in $[A, 2A)$ with $T \ge 36$, let $H = \lfloor 28a/(AT)\rfloor + 1$ and $X = \lfloor 3609600A^4H^6/a^3\rfloor + 1$. Then*

$$T^3 \le C_\star\, \Xi_2(a)^9\,(H^3X^{14} + X^{15}), \qquad C_\star = 32\cdot 607461312^3,$$

and $T^{152} \le C_{152}\Delta^{39}\Xi_2(a)^{117}a^{45}$ with $C_{152} = 122449^{38} + c_0^{13} + 8^{31}c_A + 8^{30}c_B$, where $c_0 = 1024\,C_\star\,56^3$, $c_A = 2^{15}C_\star\,3609600^{14}56^{87}$ and $c_B = 2^{16}C_\star\,3609600^{15}56^{90}$.

Proof. Let $\mathcal{S}$ be the set of minor vectors of good starts and $\tau_0 = \tau^*(P_2(a))$. At most four good starts share (p,κ), and $T \le 2|\mathcal{G}|$, so Proposition 14.7 gives

$$T \le 8\,\Xi_2(a)\Big(8\sum_{p\in\mathcal{S}}\lfloor 3H/\lambda(p)\rfloor + 3|\mathcal{S}|\Big).$$

Choose t with $X \le t^3 \le 8X$. Lemma 14.2 gives $\lambda(p) \le t$, and Proposition 14.5 and Lemma 14.8 give $t\sum_p\lfloor 3H/\lambda(p)\rfloor \le 3H|\mathcal{S}| + 3H\cdot 393660X^4\tau_0^2t^3$, with $|\mathcal{S}| \le 14581X^5\tau_0$. Hence $Tt \le 607461312\,\Xi_2^3X^5(H+t)$, and cubing, with $(H+t)^3 \le 4(H^3+t^3)$ and $X \le t^3 \le 8X$, gives the first bound.

For the second, if $AT > 28a$ then $T^4 \le 122449a$. Otherwise $H\,AT \le 56a$. If $X = 1$ then $T^6A^3 \le 2C_\star\,56^3\Xi_2^9a^3$, and $T^{12} \le (8A\Delta)^3T^6 \le c_0\Delta^3\Xi_2^9a^3$. If $X \ge 2$, write $X = X_0 + 1 \le 2X_0$ with $X_0a^3 \le 3609600A^4H^6$. When $2X_0 \le H^3$, $T^3a^{42} \ll \Xi_2^9A^{56}H^{87}$, so $T^{90}A^{31} \le c_A\Xi_2^9a^{45}$ and $T^{152} \le (8A\Delta)^{31}T^{90}$. When $2X_0 > H^3$, $T^3a^{45} \ll \Xi_2^9A^{60}H^{90}$, so $T^{93}A^{30} \le c_B\Xi_2^9a^{45}$ and $T^{153} \le (8A\Delta)^{30}T^{93}$. □

Proof of Theorem 1.8. Since $\Delta, \Xi_2(a) \ll_\eta a^\eta$, every window has $T \ll a^{(45+156\eta)/152}$. Summing over the $O(\log a)$ windows proves the theorem. □

Remark 14.10. In terms of the scales, $T \ll a^\eta(HX^{14/3} + X^5)$: the factor $X^{1/3}$ saved over Proposition 13.4 is the typical $\lambda(p)$. At the critical point $T \approx a^{45/152}$, $Q \approx a^{62/152}$, $H \approx a^{17/152}$ and $X \approx a^{6/152}$, and the term $X^5 \approx a^{30/152}$ is smaller.

15. Joint integer invariants and the optimized bound

We now use the fixed dividend in the equations of the pattern curve and adjoin small integer invariants as coordinates. This gives the strongest bound of the paper without the weighted lattice count of Section 14. Throughout this section constants may depend on fixed positive integers m, n and on a positive loss η, but never on the dividend, orbit or pattern. The argument has three steps: construct the curves for a fixed pattern, increase their degree by adjoining integer invariants, and apply the curve-counting theorem using the heights of those invariants.

15.1. **Selected windows and their curve components.** Use the remainder window $[A, 2A)$ and its number T of visits. For $T \ge 36$ set

$$H = \left\lfloor \frac{28a}{AT} \right\rfloor + 1, \qquad X = \left\lfloor \frac{3609600A^4H^6}{a^3} \right\rfloor,$$
$$S = \left\lfloor \frac{4352A^3H^3}{a^2} \right\rfloor, \qquad Y = \left\lfloor \frac{1259991040A^5H^{10}}{a^4} \right\rfloor.$$

At least $T/2$ starts have nine available states in the window and their next seven gaps at most H. These are the selected starts. Their complete minor vectors have height at most X, and $X \geq 1$. The three free node coordinates $q_{i+j} - q_i$, $j = 1, 2, 3$, are integers in $[0, 3H]$. The pattern count from Section 13 is $O_\eta(a^\eta X^5)$. Thus H controls gaps, X controls patterns, and S, Y will control the additional integer coordinates.

For the pattern $\theta = (N_0, N_1, N_2, u_1, u_2, v_1, v_2)$, put

$$A_0 = N_0 = I_4(i), \qquad (\rho, u, v) = \frac{(N_1, u_1, u_2)}{N_0}, \qquad (\rho', u', v') = \frac{(N_2, v_1, v_2)}{N_1}.$$

Here A_0 is a minor, distinct from the window scale A, and $\rho > 0$, $u > 0$, $v < 0$ by Lemma 12.1. Use the three free nodes s_1, s_2, s_3 as coordinates and define

$$s_0 = 0, \quad s_4 = (\rho + u)s_1 - (u - v)s_2 - (v - 1)s_3, \quad k_j = s_{j+1} - s_j \quad (0 \leq j \leq 3),$$
$$k_4 = \rho' k_1 - u' k_2 - v' k_3, \qquad s_5 = s_4 + k_4.$$

The first elimination is equivalent to $\rho k_0 = u k_1 + v k_2 + k_3$. Let

$$\gamma = (\rho, -\rho - u, u - v, v - 1, 1),$$

and let γ' be the same expression in the primed quantities. The orthogonal quartics of the current and next states, in the same variable, are

$$P_4(x) = \sum_{j=0}^{4} \gamma_j \prod_{\ell=0}^{j-1} (x - s_\ell), \qquad R_4(x) = \sum_{j=0}^{4} \gamma'_j \prod_{\ell=1}^{j} (x - s_\ell).$$

Put $Z = xR_4$ and $f = \sum_{j=0}^{4} \gamma_j (s_j^2 + k_j)$. The next-state quadratic relation eliminates k_5, giving the cubic polynomial

$$D = \sum_{j=0}^{4} \gamma_j \big(s_j^3 + 3s_j^2 + 3s_j(1 + k_j) + (1 + k_j)^2\big) + \sum_{j=0}^{3} \gamma_j k_{j+1}$$
$$+ \rho' r_1 - u' r_2 - v' r_3 - k_4(2s_4 + k_4),$$

where $r_j = k_j(s_j + s_{j+1}) + k_{j+1}$ for $j = 1, 2, 3$. At selected starts, $D = \nu_i(P_4^2)$ is the quartic norm. Actual points satisfy

$$f = 0, \qquad A_0 \operatorname{Res}(P_4, Z) = aD^4. \tag{12}$$

Here f is the pattern quadratic, $\deg D \leq 3$, and the second equation has degree at most twelve. The two equations are relatively prime; the quadratic has irreducible quadratic part with nonzero Hessian determinant. Their common zero set admits a cover by at most twenty-four integral affine curves over $\mathbb{Q}$, each of degree at most twenty-four. Here "integral" means that the defining ideal is prime over $\mathbb{Q}$; it does not assert geometric irreducibility or a rational parametrization. Degrees throughout this section mean the leading coefficient of the eventual affine Hilbert function, equivalently the degree of the projective closure.

15.2. **Reconstruction and component selection.** Two further identities distinguish components on which the later rational functions have the required behavior. We give their definitions before stating the component bounds.

Form the 8×9 matrix of coefficients of $P_4, xP_4, x^2P_4, x^3P_4, Z, xZ, x^2Z, x^3Z$ in the basis $1, x, \ldots, x^8$. Let c_j be $(-1)^j$ times the minor obtained by deleting column j, and define

$$Q = \frac{c_1 - c_2}{c_1 - c_0}, \qquad E_0 = a\,\frac{\det(c_{r+s})_{0 \leq r,s < 3} (c_1 - c_0)^3}{(c_1^2 - c_0 c_2)^3}.$$

At an actual point the matrix annihilates the moment vector $(h_i(0),\dots,h_i(8))$ and has rank eight. Indeed, the resultant in (12) is positive, so P_4 and Z are coprime. The identities $q_ih_i(0)+h_i(1)=q_ih_i(1)+h_i(2)=1$, together with $h_i(0)h_i(2)>h_i(1)^2$, then give

$$h_i(j)=\frac{c_j(c_1-c_0)}{c_1^2-c_0c_2}\quad(0\le j\le 8),\qquad Q=q_i,\qquad E_0=I_3(i).$$

In particular, all displayed denominators are nonzero at selected starts.

To state the identities as equations on the curve, use the raw-moment polynomials

$$M_0=1,\qquad M_{r+1}(k_0,k_1,\dots)=\sum_{t=0}^{r}\binom{r}{t}k_0^{r-t}M_t(k_1,k_2,\dots).$$

Only $M_0,\dots,M_6$ are needed; they involve $k_0,\dots,k_4$. At selected starts, $M_r=\mu_i(x^r)$ for $0\le r\le 6$, by the weight recursion in Section 10. Set

$$H_3=\det(M_{r+s})_{0\le r,s<3},\qquad H_4=\det(M_{r+s})_{0\le r,s<4},\qquad H_{013}=\det(M_{r+s})_{\substack{r=0,1,2\\ s=0,1,3}}.$$

This H_4 is the polynomial of Lemma 11.4. With $p_3=v-1-s_1-s_2-s_3$, the coefficient of x^3 in P_4, the normalization and raw-resolvent identities are

$$A_0P_4(-Q)=aH_4,\qquad E_0P_4(-Q)=a\big((Q-p_3)H_3-H_{013}\big). \tag{13}$$

Among components containing a selected start, we call a component *retained* if both equations, after clearing denominators, hold identically on it.

For clarity, the additional component facts used below are recorded together. They apply to components supporting the selected actual points, not to an assumption that every point of a component is an orbit point.

Lemma 15.1. *The following bounds are uniform over the complete minor pattern.*

1. *The exceptional patterns, characterized by a vanishing pair sum of the quartic's five Newton coefficients, number $O_\eta(a^\eta X^4)$. Outside these patterns every component has even degree: it is a conic or has degree at least four.*
2. *In the range*

$$124928A^4H^5X<a^3, \tag{14}$$

   no line component contains a selected actual point.
3. *The cleared identities* (13) *have degrees at most* 166 *and* 404*. On a prime component of degree d, either each identity holds identically or its selected actual points number at most* $166d$ *or* $404d$*, respectively.*
4. *A nonexceptional conic contributes $O_\eta(a^\eta)$ distinct gap vectors.*

Proof. Exceptional patterns and parity. The sign identities and the quadratic relation give the four possible vanishing pair sums. Each is a fixed linear condition on the minor vector; counting it in the same pivot charts as the ordinary minor count removes one power of X. On the infinity section of a nonexceptional component the leading node values are distinct. The resultant has pole order ten times the hyperplane multiplicity. The fixed-dividend equation then makes each branch's hyperplane multiplicity even; their residue-degree weighted sum is d. This proves the parity assertion, including over $\mathbb{Q}$ without a geometric irreducibility assumption.

Lines. The line equations force $aH^5(24X+464)\ge q_i^4$ at a supported point. But $q_i\ge a/(4A)$, and $X\ge 1$ makes (14) imply the strict reverse inequality. This excludes lines also for exceptional patterns.

Proper cuts. Clear the rational quotient reconstructed from the two quartics and then clear the normalization and resolvent equations. Polynomial degree bounds give 166 and 404; a

proper polynomial section of a degree-d prime curve has at most its degree times d distinct rational points. Reconstruction recovers q_i from the free nodes and the fixed pattern. Since q_i strictly increases along the orbit, distinct selected starts have distinct free-node triples. Thus these point bounds also bound the number of selected starts.

Conics. For a nonexceptional conic the double infinity section gives a polynomial parameter of degree two. One obtains an integer parameter and quadratic numerator polynomials with a common denominator, whose heights are bounded by a fixed power of a. The rank-three invariant is a proper rational function of this parameter—its numerator has smaller degree than its denominator after cancellation—except for a bounded number of points cut out by the actual normalization equations. At integral values, a polynomial Bezout identity forces the denominator value to divide a nonzero resultant of height bounded by a fixed power of a. Its number of divisors is $O_\eta(a^\eta)$; each value has boundedly many parameter preimages. Coefficient, denominator and resultant heights are uniform because the conic degrees are fixed and $X \le 10^{16}a^7$.

The formal proofs include the properness and height estimates used in this conic argument. The README maps the component, line-exclusion, parity and conic arguments to their Lean modules; the scope of the formalisation is described in Section 16. □

15.3. **The joint coordinates.** Put $d_j = b_{i+j} - b_{i+j+1}$ and

$$e_{01} = d_0k_1 - d_1k_0, \quad e_{02} = d_0k_2 - d_2k_0, \quad e_{12} = d_1k_2 - d_2k_1, \qquad W = (e_{12}, -e_{02}, e_{01}).$$

The first curvature e_{01} is $I_3(i)$. With $s_j = q_{i+j} - q_i$ and $r_j = k_j(s_j + s_{j+1}) + k_{j+1}$, write

$$B_0 = k_0r_1 - k_1r_0 = k_0^2k_1 + (k_0 - 1)k_1^2 + k_0k_2,$$
$$B_1 = k_1r_2 - k_2r_1 = k_1^2k_2 + (k_1 - 1)k_2^2 + k_1k_3.$$

They are positive at actual states, and $B_0 = H_3$. On a retained component, use the reconstructed function E_0 to define

$$E_{12} = \frac{B_1E_0 - A_0k_1}{B_0}, \qquad E_{02} = \frac{k_0E_{12} + k_2E_0}{k_1}, \qquad W = (E_{12}, -E_{02}, E_0), \qquad J = A_0D.$$

The determinant and cross-product identities give, identically,

$$W \cdot (k_0, k_1, k_2) = 0, \qquad W \cdot (r_0, r_1, r_2) = A_0. \tag{15}$$

These rational functions specialize to the displayed integer minors. Changing the rank-five Gram basis to $1, x, x^2, x^3, P_4$ gives $\det(\nu_i(x^{r+s}))_{r,s<5} = \det(\nu_i(x^{r+s}))_{r,s<4}\nu_i(P_4^2)$. Consequently $J = I_5(i)$ is a positive integer at actual states.

Lemma 15.2. *At selected actual starts, $|W_j| \le S$ and $1 \le J \le Y$.*

Proof. Use the exact positive remainder expansion $(q + y)^{-1} = q^{-1} - yq^{-2} + y^2/(q^2(q + y))$. For the first four shifted states,

$$\frac{b_{i+j}}{a} = \frac{1}{q} - \frac{s_j + 1}{q^2} + \epsilon_j, \qquad 0 \le \epsilon_j \le 17H^2/q^3.$$

Thus $d_j/a = k_j/q^2 + O(34H^2/q^3)$, and cancellation of the leading terms in each minor gives $|e_{jl}|q^3 \le 68aH^3$. The bound $q \ge a/(4A)$ gives S, including the floor because the minors are integers. For J, Newton basis Gram comparison bounds the raw rank-five determinant by $(1 \cdot 2 \cdot 10 \cdot 77 \cdot 799)H^{10} = 1230460H^{10}$. Weighted Gram comparison gives $Jq^5 \le 1230460aH^{10}$; the same lower bound for q gives Y. □

15.4. **The extra degree and the powered graph.** Let C be a retained nonexceptional component of degree $d \geq 4$. On the normalization of its projective closure, let E denote the pullback of the original hyperplane at infinity, of degree d. At a branch over infinity with hyperplane multiplicity e, the retained identities give

$$e \text{ is even}, \qquad \operatorname{ord}(J) = -5e/2, \qquad \min_j \operatorname{ord}(W_j) = 3e/2. \tag{16}$$

Here the resultant order is $-10e$, both B_0 and B_1 have order $-3e$, and H_4 has order $-6e$. Substitution in the normalization and raw-resolvent identities gives the rank-three zero order $3e/2$. The two reconstruction formulas then give the stated orders for the other curvatures. Distinct leading nodes exclude cancellation of the requisite Vandermonde factors. These calculations take place at discrete valuations of the normalized chart, so they also apply at singular points of the original curve.

Let F be the pole envelope of the three W_j: at each branch its coefficient is the maximum of their pole orders and zero. It is supported over the affine part and disjoint from E. We have

$$\deg F \geq 3d/2 + 1. \tag{17}$$

Indeed, the projective direction $[W_0 : W_1 : W_2]$ is nonconstant. Otherwise the first identity in (15) places the curve in a fixed plane, whereas a plane section of the irreducible pattern quadric has degree at most two. The second identity ensures that the curvature vector does not vanish at an affine point where its coordinates are regular. At a supported rational point choose a rational linear combination of the W_j which vanishes there but is not the zero function. On normalization it has a positive-order zero above that point, in addition to zeros of total degree at least $3d/2$ over infinity. Its poles are bounded by F. Equality of zero and pole degrees gives (17). Residue degrees are included in all these sums; the extra branch contributes at least one even when its residue field is larger than $\mathbb{Q}$.

For fixed positive integers m, n, adjoin W_0^m, W_1^m, W_2^m, J^n to the three original node coordinates. The new curve is the closure of this graph on the common domain of the rational functions. Projection onto the original coordinates is birational. A generic rational linear combination of all seven coordinates avoids cancellation at every relevant pole: each forbidden choice is a proper linear subspace over the infinite field $\mathbb{Q}$. Its old infinity poles have total degree $5nd/2$, and the affine poles contribute $m \deg F$. The degree of this rational function is at most the graph Hilbert degree δ. It follows that

$$2\delta \geq (5n + 3m)d + 2m, \qquad \delta \geq 10n + 7m =: L_*. \tag{18}$$

For $m = n = 1$ it gives $\delta \geq 4d + 1 \geq 17$. Clearing the common rational denominator gives the uniform upper bound $\delta \leq ((m+n)247+1)d$. In the unpowered case the sharper bound $247d$ also holds. Actual integer points on the powered graph lie in the box of radius $\max(3H, S^m, Y^n)$.

15.5. **Counting and the final exponents.** We use the coefficient-uniform affine-curve theorem of Castryck–Cluckers–Dittmann–Nguyen [3, Theorem 3], also stated as [4, Corollary 2.2]: an integral affine curve over $\mathbb{Q}$ of degree δ in fixed ambient dimension $N \geq 2$ satisfies

$$N_{\mathrm{aff}}(C, B) \ll_N \delta^3 B^{1/\delta}(\log B + \delta), \qquad B \geq 1. \tag{19}$$

The separate Lean library `ccdn-dimension-growth` proves the real-B version of (19), including the plane estimate, degree-preserving projection and box rescaling. It is imported as a proved theorem, with no external counting hypothesis.

For fixed m, n the graph degrees lie in a fixed interval. All the heights are bounded by fixed powers of a, so the logarithm in (19) is $O_\eta(a^\eta)$. The nonexceptional conics and proper cuts contribute $O_\eta(a^\eta X^5)$. Exceptional patterns number $O_\eta(a^\eta X^4)$; after line exclusion the

original three-coordinate curve count gives $O_\eta(a^\eta H^{1/2})$ per pattern. The other components use (18). At most four actual starts share a fixed pattern and complete gap vector. Summing the at-most-twenty-four components, allowing overlaps, gives, under the line-exclusion condition (14),

$$T \ll_{m,n,\eta} a^\eta \left(X^5 + X^4 H^{1/2} + X^5 \max(H, S^m, Y^n)^{1/L_*}\right). \tag{20}$$

The added invariants introduce no additional freely counted pattern coordinates.

We now eliminate the window scale A from the count. In the range $AT \le 28a$,

$$HAT \le 56a, \quad Xa^3 \le 3609600A^4H^6, \quad T^2 \le 8\Delta A, \qquad \Delta = \max_{1\le j\le 32a^7} \tau(j).$$

If the graph term in (20) dominates, split according to which of $\max(H, S^m)$ and Y^n is larger, and call that radius $\mathcal{B}$. Raising the bound to L_* gives $T^{L_*} \le KX^{5L_*}\mathcal{B}$, where $K \ll_{m,n,\eta} a^{\eta L_*}$ includes the counting loss. This and a box estimate $\mathcal{B}T^\beta \le C_1 a^\gamma$, where C_1 may include divisor factors, imply, by raising these inequalities and cancelling positive powers of A, a,

$$T^{51L_*+\beta} \le K\, 3609600^{5L_*} 56^{30L_*} C_1 (8\Delta)^{10L_*} a^{15L_*+\gamma}.$$

The curvature bound gives $ST^3 \ll a$, while $HT^3 \le 448\Delta a$ follows from the first and third window inequalities. Since $H \ge 1$ and $m \ge 1$, $\max(H, S^m) \le \max(H, S)^m$. This gives $(\beta, \gamma) = (3m, m)$ for $\mathcal{B} = \max(H, S^m)$, up to divisor factors. The rank-five bound gives $YT^{20} \ll \Delta^5 a^6$, hence $(\beta, \gamma) = (20n, 6n)$ for $\mathcal{B} = Y^n$. For comparison, the ordinary-pattern term gives $T^{51} \ll a^{15+O(\eta)}$, the exceptional-pattern term gives $T^{85} \ll a^{25+O(\eta)}$, and failure of (14) gives $T^{17} \ll a^{5+O(\eta)}$, after absorbing divisor factors. These exponents are all $5/17$ up to the arbitrarily small loss. The remaining range $AT > 28a$ gives $T^4 \ll a$. Thus every window, and then their dyadic sum, has exponent

$$\max\left(\frac{15L_* + m}{51L_* + 3m}, \frac{15L_* + 6n}{51L_* + 20n}\right) + \varepsilon.$$

Both displayed fractions exceed $5/17$. All divisor and logarithmic losses are absorbed after fixing m, n.

Proof of Theorem 1.1. For $m = n = 1$ the larger fraction is $128/435$. For $m = 50, n = 51$ it is $2201/7480$. In general set $r = n/m$. The two fractions become

$$E_S(r) = \frac{150r + 106}{510r + 360}, \qquad E_Y(r) = \frac{156r + 105}{530r + 357}.$$

The first decreases and the second increases on $r > 0$. Their crossing satisfies $30r^2 - 10r - 21 = 0$, so $r_* = (5+\sqrt{655})/30$ and their common value is $\alpha_* = (1310 - \sqrt{655})/4365$. Positive rational ratios approximate r_*; choose the fixed positive integers m, n to make the larger exponent less than $\alpha_* + \varepsilon/2$. Apply the proved fixed-power bound with loss $\varepsilon/2$. The powers and all constants depend only on ε, not on a, b. □

The proof is assembled in `PoweredJointBound`. Its main ingredients are the rational parity cover, ordinary and conic gap counts, graph-degree lower bounds, powered pattern count, and window estimates. The repository README maps these steps to their Lean modules. The public statements in `JointLiftMain` are `P_le_128_435`, `P_le_2201_7480` and `P_le_joint_optimal`. Their transitive proof dependencies are audited to exclude the weighted-lattice counting modules.

16. The Lean formalisation

The accompanying Lean 4 project uses toolchain `v4.35.0` and Mathlib `v4.35.0`. It proves the main results of this paper, including the curve-counting input to Theorem 1.1. The logarithmic refinement is the one exception: its Lean proof takes the two published analytic estimates (9) and (10) as explicit hypotheses.

The strongest bound has the following statement:

```
theorem P_le_joint_optimal :
    ∀ ε > (0 : ℝ), ∃ C > 0, ∀ a b : ℕ,
    0 < b → b < a → (P a b : ℝ) ≤
    C * (a : ℝ) ^ (((1310 : ℝ) - Real.sqrt 655) / 4365 + ε)
```

The order of the quantifiers makes the uniformity explicit: C may depend on ε, but not on a or b. Here `P a b` is defined by well-founded recursion,

$$P(a,0) = 0, \qquad P(a,b) = 1 + P(a, a \bmod b) \quad (b > 0),$$

so it counts exactly the positive terms of the orbit. The formal orbit is indexed from zero, as in Section 10.

The following table gives the public declarations; namespaces and the `.lean` extension are omitted. The repository README gives the fuller map from proof steps to modules.

Result	Declaration	Module
Theorem 1.1, α_*	`P_le_joint_optimal`	`JointLiftMain`
Theorem 1.1, 2201/7480	`P_le_2201_7480`	`JointLiftMain`
Theorem 1.1, 128/435	`P_le_128_435`	`JointLiftMain`
Theorem 1.2	`P_le`	`Main`
Theorem 1.5	`P_le_23_77`	`TwentyThreeMain`
Theorem 1.6	`P_le_11_37`	`ElevenMain`
Theorem 1.7	`P_le_8_27`	`EightMain`
Theorem 1.8	`P_le_45_152`	`FortyFiveMain`
Theorem 1.3	`P_le_loglog`	`Hooley`
Theorem 1.4	`P_le_rises`	`RisesMain`
Theorem 10.2	`hankelInv_pos_int`	`Hankel`

The identifications of the first four invariants in Theorem 10.2 are proved in the same module as `hankelInv_one` through `hankelInv_four`.

For the curve theorem, the separate library `ccdn-dimension-growth` proves the plane estimate, degree-preserving projection and box rescaling. The Pierce development applies this theorem to the defining prime ideals of its curves, using the same affine Hilbert degree as in Section 15. Thus the curve count is a proved dependency, not a hypothesis of the Pierce bound. In contrast, `P_le_loglog` takes the propositions `NairTenenbaumDelta` and `BretecheTenenbaumMean`, which state (9) (with $\sigma = 1/4$) and (10), including their existential constants and thresholds.

The file `Axioms.lean` audits the eleven public statements and their proof dependencies. The only logical axioms reported are `propext`, `Classical.choice` and `Quot.sound`, the standard axioms used by Mathlib. The proof library contains no `sorry` and uses no `native_decide`. The separate Mathlib-only `Challenge.lean` states the eleven results with deliberate theorem holes for comparison with the proved `Solution.lean`; neither module imports the other. Build instructions and the scope of the verification checks are recorded in the README.

Many local identities are checked by polynomial normalization over $\mathbb{Z}$, then transferred to the natural-number orbit after proving that the relevant differences are nonnegative. The counting arguments bound the number of parameter values and then the number of states

with each value. Determinant positivity, integrality, curve reconstruction and the valuation arguments are proved within the same development. The divisor estimate in Lemma 2.2 uses `DivisorPowerBound.lean` from `openai/math` [10] (Apache-2.0), with unchanged proof bodies and adapted imports and module visibility. The required Hadamard inequality is proved in `Hadamard.lean`.

17. Limits of the method

The joint-invariant bound of Section 15 has a remaining threshold $5/17$ from the ordinary pattern and complementary-scale terms. Improving the coordinate embedding alone does not remove those contributions. A stronger curvature degree bound or a sharper count of realizable patterns may improve on α_*. No claim of optimality for the Pierce iteration is made.

Proposition 7.3 removes the divisor loss from the local bound, but on its own the exponent is still $3/10$. The local bound $T \ll a^{1/2}A^{-1/3}$ and the drop bound $T \ll A^{1/2}$ (up to the divisor factor) both allow $T \asymp a^{3/10}$ at $A \asymp a^{3/5}$. Even an ideal short-interval divisor estimate with G constant leaves this balance unchanged. So sharpening constants or improving the divisor counts in that argument cannot give a logarithmic bound. To improve that argument, one needs to restrict *which* configurations of the parameters (u, v, w, z, D, E, F) can occur together along a single orbit. Theorem 1.4 is one restriction of this kind: a long orbit must have many rises in its quotient increments. Theorem 10.2 supplies the natural candidates for further ones, the higher invariants I_r. Section 11 uses I_4 in exactly this way.

The count of Section 12 loses through the number of patterns, X^7, reduced to X^5 in Section 13 and weighted by the lattices Λ_p in Section 14. With patterns at four consecutive states one has thirteen integers and as many equations as unknown increments, so one might hope for $O(1)$ configurations per pattern. But the four-state equations have positive-dimensional families of solutions. For instance, the Newton coefficients $(\alpha, -3\alpha-1, 3\alpha+3, -\alpha-3, 1)$, with coefficient polynomial $(z-1)^3(z-\alpha)$, at every state, together with constant increments $k_j = t$, satisfy all the linear and quadratic relations for every t. Such families are excluded only by the remainder equations, which those earlier counts discard. The fixed-dividend curve equations of Section 15 retain this information. Section 14 uses only the first minimum of Λ_p, and slices in a single direction; the other two minima and the shape of the quadric in the lattice are not used.

There is no obvious monotonicity to exploit. The curvatures D_i of successive blocks need not decrease. For $a = 13242$ the orbit of 255 begins

$$255,\ 237,\ 207,\ 201,\ 177,\ 144,\ 138,\ 132,\ \ldots$$

with quotients $51, 55, 63, 65, 74, 91, 95, 100, \ldots$. The curvatures of the first five blocks are $24, 12, 6, 111, 30$, all for blocks that lie within a factor of two.

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application of Shiu's theorem identified above. We thank the Mathlib community, and the authors of `openai/math` for the formalised divisor bound.

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